

TERRESTRIAL BIODIVERSITY IMPACT ASSESSMENT REPORT

For The

PROPOSED PHASE 1 GILIMBERG RE 1, 200MW CLUSTER SOLAR PV FACILITY AND
ASSOCIATED INFRASTRUCTURE IN MOGALAKWENA LOCAL MUNICIPALITY,
WATERBERG DISTRICT, LIMPOPO PROVINCE.



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2 Indemnity

- This report is based on survey and assessment techniques, which are limited by time and budgetary constraints relevant to the type and level of investigation undertaken.
- This report is based on a desktop and field-based investigation using available information and data related to the site to be affected, *in situ* fieldwork, surveys, assessments, and the specialist's best scientific and professional knowledge.
- The Precautionary Principle has been applied throughout this investigation.
- The findings, results, observations, conclusions, and recommendations given in this report are based on the specialist's best scientific and professional knowledge as well as information available at the time of the study.
- Additional information may become known or available during a later stage of the process for which no allowance could have been made at the time of this report.
- The specialist reserves the right to modify this report, recommendations, and conclusions at any stage should additional information become available.
- Information and recommendations in this report cannot be applied to any other area without proper investigation.
- This report, in its entirety or any portion thereof, may not be altered in any manner or form or for any purpose without the specific and written consent of the specialist as specified above.
- Acceptance of this report, in any physical or digital form, serves to confirm acknowledgment of these terms and liabilities.

Lead Specialist: Rebotile Rachuene

3 Declaration of Independence

3.1 First Specialist

I, Mr Rebotile RACHUENE, registered as a Professional Natural Scientist, in my capacity as a specialist consultant, hereby declare that I:

- Act/Acted as an independent specialist in this project.
- Do not have any personal, business, or financial interest in the project except for financial remuneration for specialist investigations completed in my professional capacity as specified by the Environmental Impact Assessment Regulations, 2014, as amended.
- Will not be affected by the outcome of the environmental process, of which this report forms part of.
- Do not have any influence over the decisions made by the competent authorities.
- Do not object to or endorse the proposed developments but aim to present facts and my best scientific and professional opinion about the impacts of the development.

Undertake to disclose to the relevant authorities any information that has or may have the potential to influence its decision or the objectivity of any report, plan or document required in terms of the Environmental Impact Assessment Regulations, 2014, as amended.

Rebotile Rachuene



20 April 2026

3.2 Second Specialist

I, Mr Prince Molokomme, registered as a Candidate Natural Scientist, in my capacity as a specialist consultant, hereby declare that I:

- Act/Acted as an independent specialist in this project.
- Do not have any personal, business, or financial interest in the project except for financial remuneration for specialist investigations completed in my professional capacity as specified by the Environmental Impact Assessment Regulations, 2014, as amended.
- Will not be affected by the outcome of the environmental process, of which this report forms part of.
- Do not have any influence over the decisions made by the competent authorities.
- Do not object to or endorse the proposed developments but aim to present facts and my best scientific and professional opinion about the impacts of the development.

Undertake to disclose to the relevant authorities any information that has or may have the potential to influence its decision or the objectivity of any report, plan or document required in terms of the Environmental Impact Assessment Regulations, 2014, as amended.

Prince Molokomme



20 April 2026

4 EXECUTIVE SUMMARY

Sediba Sa Metse (Pty) Ltd was appointed to undertake a Terrestrial Biodiversity Impact Assessment (TBIA) for the proposed Phase 1 Gilimberg RE 1, 200MW Cluster Solar PV Facility and associated infrastructure, including its 2km powerline, located approximately 2km east of Mars Village in the Mogalakwena Local Municipality, Waterberg District, Limpopo Province. The assessment, at the 300ha, was conducted in accordance with relevant environmental legislation, including the National Environmental Management Act (NEMA), the National Environmental Management: Biodiversity Act (NEMBA), the National Water Act (NWA), and Environmental Impact Assessment Regulations, 2014 (Amendment G. 49028, No.3717 of July 2023 and G. 43855, No. 1150 of October 2020) as in NEMA No. 107 of 1998 as per Sections 24(5)(a) and (h) and 44, and followed applicable best-practice guidelines.

The study combined a desktop assessment using spatial biodiversity datasets (SANBI BGIS, NBA, POSA, and Red List data) with a field survey conducted from 12–16 March 2026. The aim was to determine the site's ecological sensitivity, identify Species of Conservation Concern (SCC), assess potential impacts, and recommend appropriate mitigation measures. The project area is characterised as a semi-natural system, historically used for game farming and subsistence agriculture, and currently utilised for communal livestock grazing, with additional disturbances including firewood collection and localised sand mining. Despite these pressures, the vegetation remains largely intact, with approximately 95 plant species recorded, predominantly indigenous, and 92% is dominated by indigenous. There are two protected species on site. Protected species, Sensitive Plant 1252 and the Marula tree (*Sclerocarya birrea afra*) have been observed on site.

The site falls within the Makhado Sweet Bushveld (Vulnerable) and Polokwane Plateau Bushveld vegetation types. Although the project area is classified as CBA 2, it does not fall within protected areas, Key Biodiversity Areas (KBA), or threatened ecosystems, and no major aquatic systems are present apart from ephemeral drainage lines around it.

The assessment confirmed the presence of one protected plant species and identified a high likelihood of occurrence of additional SCC, including *Elaeodendron transvaalense* (which is Near Threatened). The site supports moderate biodiversity and retains ecological functionality, including habitat provision and local connectivity and corridors.

The ecological sensitivity of the site is therefore rated as medium, with a Moderate Present Ecological Status (PES) and Ecological Importance and Sensitivity (EIS).

Potential impacts associated with the proposed solar development include:

- Localised permanent habitat loss and vegetation clearing
- Disturbance to the distribution and movement of flora and fauna
- Soil erosion and sedimentation
- Infestation and spread of invasive alien plant species

- Minor to moderate alteration of ecological connectivity/corridor and surface hydrology

Impact assessment results indicate that pre-mitigation impacts range from moderate to high, particularly during the construction phase. However, with the implementation of appropriate mitigation measures, all impacts can be reduced to low significance.

Key mitigation measures include:

- Application of the mitigation hierarchy is recommended.
 - This prioritises the avoidance and minimising of impacts of all drainage lines and the floral SCC.
- Pre -construction botanical walk to identify all the SCC locations and quantity must be undertaken (before vegetation clearance). Implementation of search and rescue where avoidance is not possible.
- All the SCC plant species must be quantified and reasonable numbers of them must be relocated to the suitable habitat that is identified and not to be destroyed.
- Micro-siting of infrastructure, such as powerline pylons, battery storage and others must avoid sensitive features
- Protection of semi-natural vegetation and drainage features with at least 50m buffer
- Implementation of erosion control and stormwater management
- Establishment of an Alien Invasive Plant Control Programme and Rehabilitation Plan
- The rehabilitation of all areas destroyed during this project must prioritise the use of indigenous species
- A waste management plan that must prioritise avoidance, minimising, prevention and protection of the environment must be developed and adopted. All persons working in the plant must be educated/inducted on this. The informative and awareness boards/posters/leaflets must be clearly displayed in publicly accessible areas.
- Ongoing monitoring and compliance by an Environmental Control Officer (ECO) throughout the whole project life span.

Residual impacts are expected to be low and localised, and cumulative impacts are considered low to moderate within the broader landscape context, considering the wider area is disturbed by residential (houses/villages) and the associated activities such as firewood collection, livestock/grazing.

Based on the findings of the assessment and specialist judgement, the Phase 1 Gilimberg RE 1, 200MW Cluster Solar PV Facility and associated infrastructure, including its 2km powerline, is considered environmentally acceptable and feasible, provided that all recommended mitigation measures are implemented, monitored, and enforced throughout the project lifecycle.

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8 LIST OF ACRONYMS & ABBREVIATIONS

AIP	Alien Invasive Plant
BGIS	Biodiversity Geographic Information System
CBA	Critical Biodiversity Areas
CMS	Conservation of Migratory Species
CR PE	Critically Endangered (Possibly Extinct
CR	Critically Endangered
DDD	Data Deficient— Insufficient Information
DMRE	Department of Mineral Resources and Energy
DWS	Department of Water and Sanitation
EAA	Environmental Authorisation Application
EAP	Environmental Assessment Practitioners
EC	Ecological Categories
EIS	Ecological Importance and Sensitivity
EN	Endangered
EW	Extinct in the Wild
EX	Extinct
GA	General Authorisation
GN R	General Regulation
GPS	Global Positioning System
HGM	Hydrogeomorphic unit
IBAs	Important Biodiversity Areas
IUCN	International Union for Conservation of Nature
QGIS	Quantum Geographic Information System
LC	Least Concern
MPRDA	Minerals and Petroleum Resources Development Act
NBA	National Biodiversity Assessment
NEMA	National Environmental Management Act
NEMBA	National Environmental Management: Biodiversity Act
NEMWA	Waste Management Act

NFEPA	National Freshwater Ecosystem Priority Areas
NT	Near Threatened
NWA	National Water Act
PCDs	Pollution Control Dams
PES	Present Ecological Status
POSA	Plants of South Africa
RAM	Risk Assessment Matrix
RDL	Red Data Listed
ROM	Run-of-mine
SACNASP	South African Council for Natural Scientific Professionals
SCC	Species of Conservation Concern
SR	Significance Rating
TOPS	Threatened or Protected Species
VU	Vulnerable
WULA	Water Use Licence Application

9 INTRODUCTION

Abantu Environmental Services appointed Sediba Sa Metse (Pty) Ltd to undertake a Terrestrial Biodiversity Impact Assessment for the basic assessment process for a Phase 1 Gilimberg RE 1, 200MW Cluster Solar PV Facility and associated infrastructure, including its 2km overhead power line connecting to the Boruto Eskom substation. The proposed 300ha site is located approximately 30km north of Mokopane, outside Sekuruwe village, in the Mogalakwena Local Municipality, Waterberg District, Limpopo Province. The proposed development of a solar facility, by its nature, has a direct impact on birds.

A team of biodiversity specialists was dispatched to the site, conducted a field assessment, and collected necessary data. The evaluation aimed to assess the area's biodiversity value and importance, and to identify key biological diversity. The assessment uses a desktop (including a literature review) and a field survey. The field assessments were undertaken between the 12th and 16th of March 2026. The assessment aimed to assess the area's biodiversity value and importance, and to identify key biological diversity. This detailed report presents the results of the investigation of the site and its surroundings.

This specialist terrestrial/biodiversity impact assessment study was undertaken in line with the Department of Water and Sanitation (DWS) Best Practice Guideline for Impact Prediction and is guided by following legislative requirements: National Environmental Management Act, 1998 (Act No. 107 of 1998) (NEMA) and Environmental Impact Assessment Regulations, 2014 (Amendment G. 49028, No.3717 of July 2023 and G. 43855, No. 1150 of October 2020, as in NEMA, Sections 24(5)(a) and (h) and 44; Regulation 636 under the National Environmental Management: Waste Act; National Water Act (Act 36 of 1998) (NWA); and NWA amendment of Regulation 704 (GN R 704) of 1999.

9.1 Terms of Reference

The terms of reference for this study were set and are as follows:

- Identify, assess, and delineate any biodiversity significance of sites, areas, or features in and around the project site.
- Search and identify any Species of Conservation Concern (SCC) and driver species.
- Assess current impacts and suggest mitigation measures for minimising impacts,

- Compile a detailed report with maps and recommendations.

9.2 Scope of Work

The following tasks were undertaken in fulfillment of the terms of reference for this assessment:

- Conduct a biodiversity desktop and field-based assessment for the proposed study
- Provide the evaluation of the area's conservation importance and significance,
- Determine sensitivity areas (maps) around the site.
- Provide detailed potential impacts from the project and
- Produce a detailed report with findings and recommendations.

9.3 Project Area of Influence (PAOI)

The **Project Area of Influence (PAOI)** includes all areas that may be affected directly or indirectly by the proposed development.

For the Phase 1 Gilimberg Solar Farm, the PAOI includes:

- The development footprint, including solar panels, inverters, and associated infrastructure;
- All access roads, cabling routes, and powerline connections;
 - Areas affected by indirect impacts, including:
 - Dust generation
 - Noise disturbance
 - Soil erosion and runoff
 - Alteration of natural drainage patterns

The PAOI was used to assess ecological receptors, including vegetation communities, fauna, and SCC, in line with the Species Environmental Assessment Guideline

9.4 Protocol Compliance Statement

This Terrestrial Biodiversity Impact Assessment has been conducted in accordance with:

- The Terrestrial Biodiversity Protocol (2020);
- The Terrestrial Plant and Animal Species Protocols (2020);
- The Species Environmental Assessment Guideline (2022);

The assessment included:

- Use of the Environmental Screening Tool and verification of sensitivity;
- A desktop assessment using national biodiversity datasets (SANBI BGIS, Red Lists, POSA);

- A field-based assessment to confirm habitat condition and species presence;
- Identification and assessment of Species of Conservation Concern (SCC);
- Evaluation of ecological sensitivity, impacts, and mitigation measures in accordance with protocol requirements.

10 APPLICABLE LEGISLATIVE AND ADMINISTRATIVE FRAMEWORK

The biodiversity assessment was undertaken in accordance with the environmental and biodiversity legislation applicable to the project, as well as relevant provincial regulations and international agreements to which South Africa is a signatory.

The Constitution of the Republic of South Africa Act, 1996 (Act No. 108 of 1996) – Section 24.

The Constitution is South Africa's overarching law. It prescribes minimum standards that existing and new laws must meet. Chapter 2 of the Constitution contains the Bill of Rights, which enshrines fundamental human rights. The government's commitment to give effect to the environmental rights enshrined in the Constitution is evident in the enactment of various pieces of environmental legislation since 1996, including the NWA and NEMA.

National Environmental Management Act, 1998 (Act No. 107 of 1998) (NEMA), as amended.

NEMA replaces several provisions of the Environment Conservation Act, 1989 (Act No. 73 of 1989) (ECA). The Act provides for cooperative environmental governance by establishing principles for decision-making on environmental matters, institutions to promote cooperative governance, and procedures for coordinating ecological functions. The principles enshrined in NEMA guide the interpretation, administration, and implementation of the Act regarding the protection and/or management of the environment. These principles serve as a framework for formulating environmental management. Section 2(4) specifies that "sustainable development requires the consideration of all relevant factors, including aspects specifically relevant to biodiversity".

National Environmental Management: Biodiversity Act, 2004 (Act No. 10 of 2004) (NEMBA).

NEMBA provides for the management and conservation of biological diversity and its components; the use of indigenous biological resources in a sustainable manner; the fair and equitable sharing of benefits arising from the bioprospecting of biological resources; and cooperative governance in biodiversity management and conservation within the framework of NEMA.

National Water Act, 1998 (Act No. 36 of 1998) (NWA).

The NWA is a legal framework for the effective and sustainable management of water resources in South Africa. Central to the NWA is the recognition that water is a scarce resource in the country, which belongs to all the people of South Africa and needs to be managed sustainably to benefit all members of society. The NWA places a strong emphasis on protecting water resources in South Africa, especially against exploitation, and on ensuring that there is water for social and economic development in the country for present and future generations.

The NWA requires any development to secure Water Use Licenses (WULs) for the following activities that apply to this project:

Section 21 (c) and (i) use, i.e., river or wetland crossings, which include any drainage lines by any infrastructure. In terms of the definitions provided, activities included under Sections 21(c) and (i) are (amongst others) for the construction of roads, bridges, pipelines, culverts, and structures for slope stabilisation and erosion protection.

General Authorisation in terms of section 39 of the NWA

According to the preamble to Part 6 of the NWA, “This Part established a procedure to enable a responsible authority, after public consultation, to permit the use of water by publishing general authorisations in the Gazette...” “The use of water under a general authorisation does not require a license until the general authorisation is revoked, in which case licensing will be necessary...”

The General Authorisations for Section 21 (c) and (i) water uses (impeding or diverting flow or changing the bed, banks, or characteristics of a watercourse) as defined under the NWA have recently been revised (Government Notice R509 of 2016). Determining whether a WUL is required for these water uses now involves the risk of degrading the ecological status of a watercourse. A low risk of impact could be authorised under General Authorisations (GAs).

Government Notice 509 of 2016

Authorisations related to wetlands are regulated by Government Notice 509 of 2016 regarding Section 21(c) and (i). This notice grants a General Authorisation (GA) for water-related uses under certain conditions, such as that all water uses must be regulated and registered with the relevant authority. The Notice sets out the conditions and considerations that should be taken into account. For instance, the user of the water must ensure that the selection of a site for establishing any impeding or diverting the flow or altering the bed, banks, course or characteristics of the watercourse works: (i) is not located on a bend in the watercourse; (ii) avoid high gradient areas, unstable slopes, actively eroding banks, interflow zones, springs, and seeps.

Convention on the Conservation of Migratory Species (CMS)

The Convention on the Conservation of Migratory Species of Wild Animals (also known as CMS or Bonn Convention) aims to conserve terrestrial, aquatic, and avian migratory species throughout their range. It is an intergovernmental treaty, concluded under the auspices of the United Nations Environment Programme, that concerns the conservation of wildlife and habitats on a global scale. Since the Convention entered into force, its membership has grown steadily to include 117 (as of 1 June 2012) Parties from Africa, Central and South America, Asia, Europe, and Oceania. South Africa is a signatory to this convention.

Provincial legislation

In addition to national legislation such as the Protected Areas Act No. 57 of 2003, the National Environmental Management: Biodiversity Act No. of 2004 and Conservation of Agricultural Resources Act No. 43 of 1983, some of South Africa's nine provinces have their own provincial biodiversity legislation, as nature conservation is a concurrent function of national and provincial government in terms of the Constitution (Act 108 of 1996).

Limpopo Environmental Management Act (LEMA)

LEMA is provincial legislation that aims to conserve biodiversity and manage natural resources in Limpopo Province. It addresses matters such as the protection of provincial nature reserves, management of protected species, and regulation of environmental activities within sensitive areas, including Critical Biodiversity Areas (CBAs). The project area falls within both CBA 1 and CBA 2, making LEMA provisions particularly relevant to development planning and biodiversity protection in this case.

Other Relevant Guidelines

- National Biodiversity Offset Guideline, 2023

The “Biodiversity Offset” in its simple terms and meaning is *“the measurable results or outcomes that are implemented to compensate nature for unavoidable impacts from development after the mitigation hierarchy measures have been exhausted”*. After the Hierarchy Mitigation (*Avoidance, Minimise, Restore and Offset*) is explored, the Offset is the last resort. These guidelines must be read together with other pieces of legislation, such as EIA (2014), NEMA (107 of 1998), and others.

11 Project Locality

The proposed 300ha project site is located at (23°54'5.85" S and 29° 1'2.35" E), approximately 2.5 km west of Mars village, about 7km east of N11, and 30km north of Mokopane town, in the Mogalakwena Local Municipality, Waterberg District, Limpopo (Figure 1).

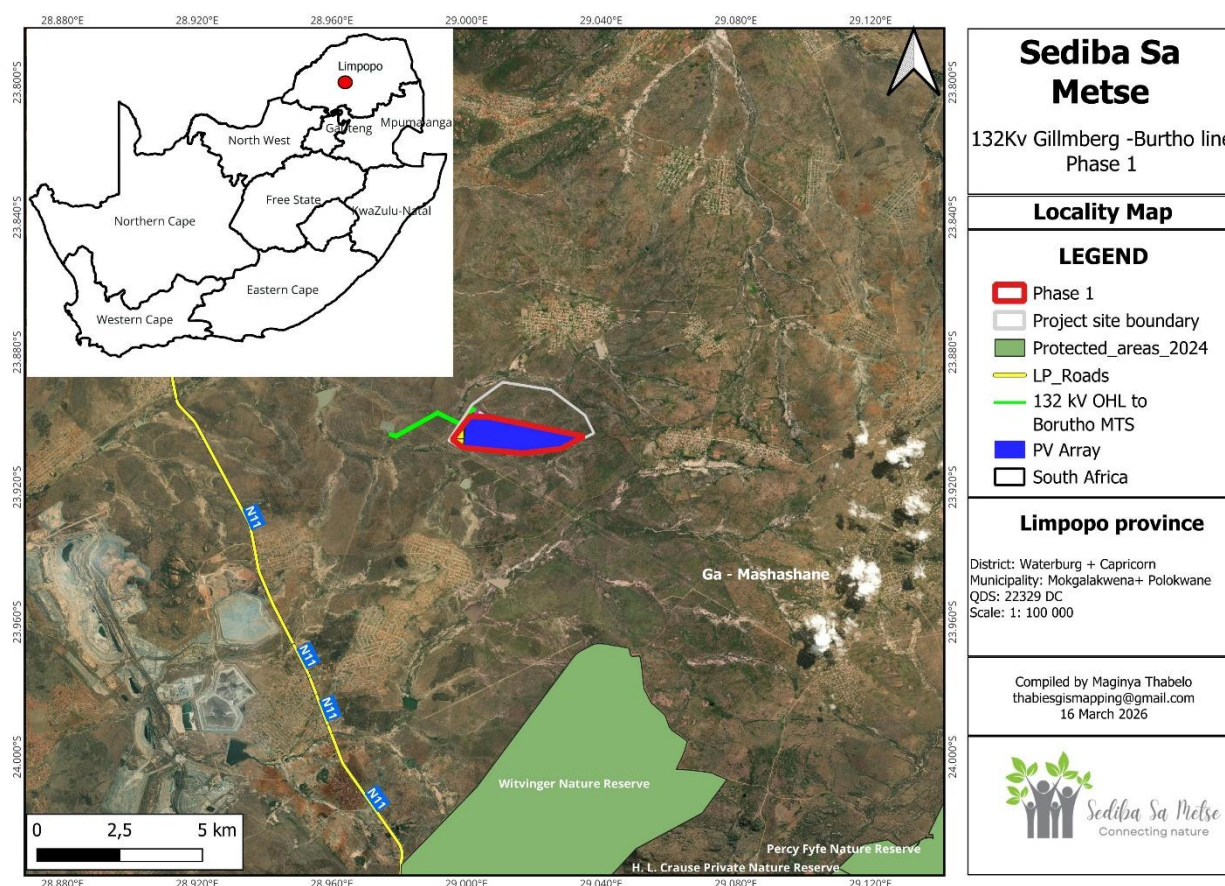


Figure 1: Locality map of the Phase 1 Gilmberg Solar PV farm.

11.1 RECEIVING ENVIRONMENT

The study area was assessed through a desktop review of available environmental data, supplemented by field surveys conducted on **12–16 March 2026**. The survey provided a snapshot of habitat conditions and terrestrial biodiversity presence within the Project Area of Influence (PAOI), with results considered sufficient to inform habitat suitability, ecological sensitivity, and the likelihood of occurrence of Species of Conservation Concern (SCC), although seasonal variation may not be fully represented.

11.2 Climate

Summer rainfall with very dry winters. MAP from about 400 mm in the northwest to about 600 mm where it borders on the foot of mountains to the east and south. Frost is fairly infrequent. Mean monthly maximum and minimum temperatures for Polokwane are 33.2°C and 0.6°C for October and June, respectively (Mucina and Rutherford, 2006).

11.3 Vegetation types and biophysical descriptions

Vegetation units are broadly classed and may include several distinct vegetation communities within a unit (Mucina and Rutherford, 2006). The vegetation type found on the study site is Makhado Sweet Bushveld and Polokwane Plateau Bushveld (Figure 2).

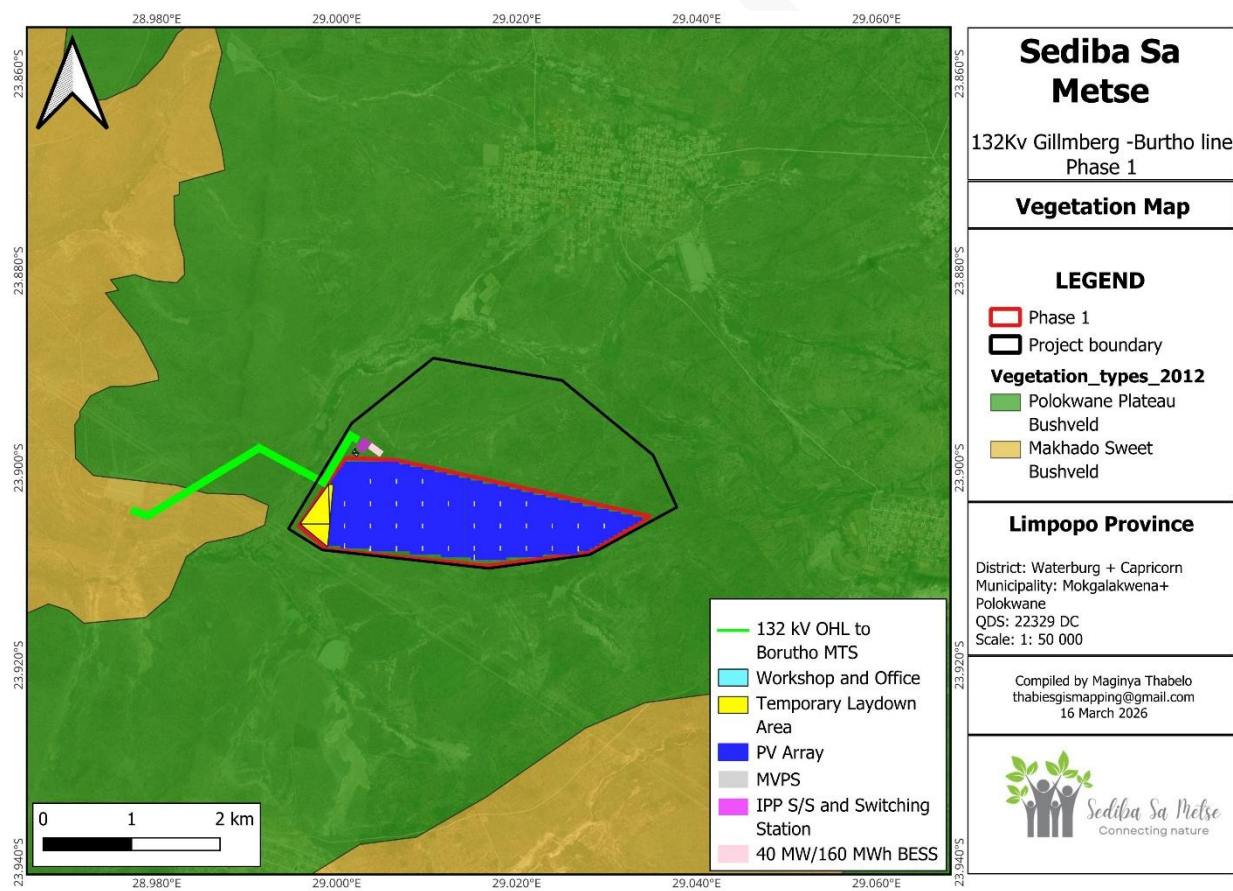


Figure 2: Vegetation Map for the Phase 1 Gilimberg Solar PV Farm.

11.3.1 Makhado Sweet Bushveld

Distribution

Limpopo Province: Straddling the Tropic of Capricorn, it occurs on the plains south of the Soutpansberg, east of the Waterberg, and on the apron surrounding the Blouberg and Lerataupje Mountains, and north of the Polokwane Plateau and west of the escarpment, with extensions to Mokopane to the south and to the north near Vivo. Altitude about 850–1 200 m.

Vegetation & Landscape Features

Slightly to moderately undulating plains sloping generally down to the north, with some hills in the southwest. Short and shrubby bushveld with a poorly developed grass layer.

Conservation

Vulnerable. Target 19%. About 1% statutorily conserved, mainly in the Bellevue Nature Reserve. Some 27% transformed, mainly by cultivation, with some urban and built-up areas. The southwestern half of the unit is home to densely populated rural communities. Erosion is low to high.

11.3.2 Polokwane Plateau Bushveld

Distribution

Limpopo Province: The higher-lying plains around Polokwane, north of the Strydpoort Mountains and south of the Makhado Sweet Bushveld. Altitude about 1,100– 1,500 m.

Vegetation & Landscape Features

Moderately undulating plains with short open tree layer with a well-developed grass layer to grass plains with occasional trees at higher altitudes (for example, the Mashashane area in the southwest and the southeastern watershed of the Sand River catchment, such as around Eersteling). Hills and low mountains of the Mamabolo Mountain Bushveld are embedded within this unit.c

Conservation

Least threatened according to remote sensing sources, but with over one-third of the remaining vegetation regarded as degraded, it would probably be regarded as susceptible. Target 19%. Less than 2% statutorily conserved mainly in the Percy Fyfe and Kuschke Nature Reserves. In addition, 0.7% conserved in other reserves, for example, the Polokwane Game Reserve. Some 17% transformed, including about 10% cultivated and 6% urban and built-up. A dense concentration of rural human settlements is found particularly in the eastern and northwestern

parts of the unit. In some regions, scattered populations of alien *Agave*, *Jacaranda mimosifolia*, *Melia azedarach*, *Opuntia ficus-indica*, and *Ricinus communis* are of concern. Erosion is high to moderate.

12 CONSERVATION STATUS

South Africa, through its various pieces of legislation, protects all biodiversity and aquatic systems. For instance, the conservation target for freshwater is at least 20%. This proposes and emphasizes that, for each freshwater ecosystem (including wetlands), at least 20% of its entire system, including its features (catchment), must be protected or conserved. Furthermore, provinces and municipalities also have their regulations, guidelines, and frameworks to conserve biodiversity ecosystems within their boundaries.

Critical Biodiversity Areas (CBAs) are terrestrial and aquatic areas of the landscape that must be maintained in a natural or near-natural state to ensure the continued existence and functioning of species and ecosystems, and the delivery of ecosystem services. Maintaining an area in a natural state can include a variety of biodiversity-compatible land uses and resource uses. It is highly recommended that this Conservation Plan serve as a primary reference for biodiversity considerations in Environmental Impact Assessments.

Criteria for Identifying CBA

Together with protected areas, the portfolio of CBAs identified in a biodiversity plan must collectively meet biodiversity targets for the representation of ecosystem types and species of special concern. These areas may also meet biodiversity targets for some ecological processes (Ecological Support Areas, ESAs) that are not essential for meeting biodiversity representation targets (thresholds), but which nevertheless play an important role in supporting the ecological functioning of critical biodiversity areas. The areas may provide ecosystem services that support socio-economic development, such as water provision, flood mitigation, or carbon sequestration (e.g., wetlands). The degree or extent of restriction on land use and resource use in these areas may be lower than that recommended for CBAs.

Criteria for Identifying ESAs

An ESA is an area that must remain in at least fair ecological condition to meet biodiversity targets for environmental processes that have not been met in CBAs or protected areas; meet biodiversity targets for the representation of ecosystem types or species of special concern when it is not possible to meet them in CBAs; support ecological functioning of a protected area or CBA (e.g. protected area buffers); or a combination of these.

ESAs can meet biodiversity targets for terrestrial, aquatic, or both. All ecological processes necessary for the long-term persistence of ecosystems and species should be adequately included in the portfolio of protected areas, CBAs, and ESAs. Sites selected to form part of an ESA could be in good, fair, or even severely modified ecological conditions, as long as the current environmental conditions are compatible with fulfilling the purpose for which the ESA has been selected. The desired state/management objective for most ESAs is to maintain them in at least fair ecological condition. For severely modified ESAs, the management objective is to prevent further deterioration of the current ecological condition.

Limpopo Conservation Plan v2 (2013)

This section of the assessment describes the biodiversity planning framework relevant to the project area, with specific reference to the Limpopo Conservation Plan version 2 (LCPv2). The Limpopo Conservation Plan v2 is the provincial biodiversity sector plan developed by the Limpopo Department of Economic Development, Environment and Tourism (LEDET) to guide biodiversity conservation and inform land-use planning and environmental decision-making within Limpopo Province.

The primary purpose of the Limpopo Conservation Plan v2 is to develop the spatial component of a bioregional plan by identifying Critical Biodiversity Areas (CBAs) and Ecological Support Areas (ESAs), together with associated land-use management guidelines. These spatial priorities represent the minimum areas required to ensure the persistence of biodiversity patterns and ecological processes in the province. The plan is intended to inform environmental impact assessments, spatial development frameworks, environmental management frameworks, and other land-use planning tools, thereby supporting sustainable development and biodiversity conservation.

Objectives of the Limpopo Conservation Plan v2

The objectives of the Limpopo Conservation Plan v2 are to:

- Identify and map Critical Biodiversity Areas and Ecological Support Areas within Limpopo Province.
- Ensure that biodiversity targets for terrestrial and aquatic ecosystems are achieved.
- Conserve representative samples of biodiversity patterns, including threatened ecosystems and endemic vegetation types.
- Maintain ecological and evolutionary processes required for long-term ecosystem functioning.

- Provide spatial biodiversity priorities and land-use management guidelines to inform planning, environmental authorisations, and resource management decisions

The conservation planning process applied a systematic, target-driven spatial biodiversity planning approach based on the best available biodiversity and spatial data, aligned with national bioregional planning guidelines.

Critical Biodiversity Areas and Ecological Support Areas

The Limpopo Conservation Plan v2 identifies biodiversity priority areas using two main spatial categories:

- Critical Biodiversity Areas (CBAs)
- Ecological Support Areas (ESAs)

Approximately 40% of Limpopo Province has been identified as Critical Biodiversity Areas, while a further 22% has been identified as Ecological Support Areas, highlighting the importance of these areas for maintaining biodiversity and ecological processes within the province

Table 1: Definitions of CBA and ESA categories used in Limpopo Province

Category	Description	Management Objective
CBA 1 (Critical Biodiversity Area 1)	Natural or near-natural areas that are considered irreplaceable for meeting biodiversity targets and ensuring the persistence of ecosystems, species, and ecological processes. These areas are essential for achieving conservation goals.	Maintain in a natural state. No habitat loss should occur.
CBA 2 (Critical Biodiversity Area 2)	Natural or near-natural areas are required to meet biodiversity targets but are considered replaceable under certain conditions. These areas contribute to achieving biodiversity targets and maintaining ecological functioning.	Maintain in a natural or near-natural state. Limited and carefully managed development may be considered, subject to environmental assessment.
ESA 1 (Ecological Support Area 1)	Natural or near-natural areas that support the functioning of CBAs and ecological processes such as water flow, species movement, and ecosystem resilience.	Maintain ecological function and connectivity. Limited disturbance may be acceptable if ecological functioning is maintained.
ESA 2 (Ecological Support Area 2)	Degraded or modified areas that are still important for supporting ecological processes or maintaining connectivity between biodiversity priority areas. These areas may require rehabilitation or careful management.	Maintain or restore ecological functioning where possible.

Protected Areas	Legally protected areas such as national parks, nature reserves, and protected environments that form part of the formal conservation network.	Maintain in accordance with conservation legislation and management plans.
Other Natural Areas (ONA)	Natural or near-natural areas that are not classified as CBAs or ESAs but still retain biodiversity value.	Avoid unnecessary habitat loss and maintain ecological integrity where possible.
No Natural Remaining Areas (NNR)	Areas that have been transformed by development, agriculture, or infrastructure and no longer retain natural habitat.	Limited biodiversity value. Restoration may be possible in some areas.

Application of the Limpopo Conservation Plan v2 to the Project Area

The Limpopo Conservation Plan v2 provides a spatial framework to guide land-use planning and biodiversity conservation within Limpopo Province. It identifies areas essential to meeting biodiversity targets and maintaining ecological processes.

Where development is proposed within areas classified as CBA 1, CBA 2, or ESA, these areas are considered environmentally sensitive and require careful management to avoid or minimise impacts on biodiversity. The presence of CBAs or ESAs within or adjacent to the proposed Citrus Farm Extension indicates that biodiversity considerations must be incorporated into project planning, design, and implementation.

The Limpopo Conservation Plan v2 is designed to inform environmental impact assessments and land-use decision-making, but it does not replace the need for site-specific ecological assessments. Detailed field verification is required to confirm the site's ecological condition and sensitivity and to guide appropriate mitigation measures. According to the Limpopo Conservation Plan version 2 (LCPv2) (2013), the study area is classified as CBA2 ([Figure 3](#)).

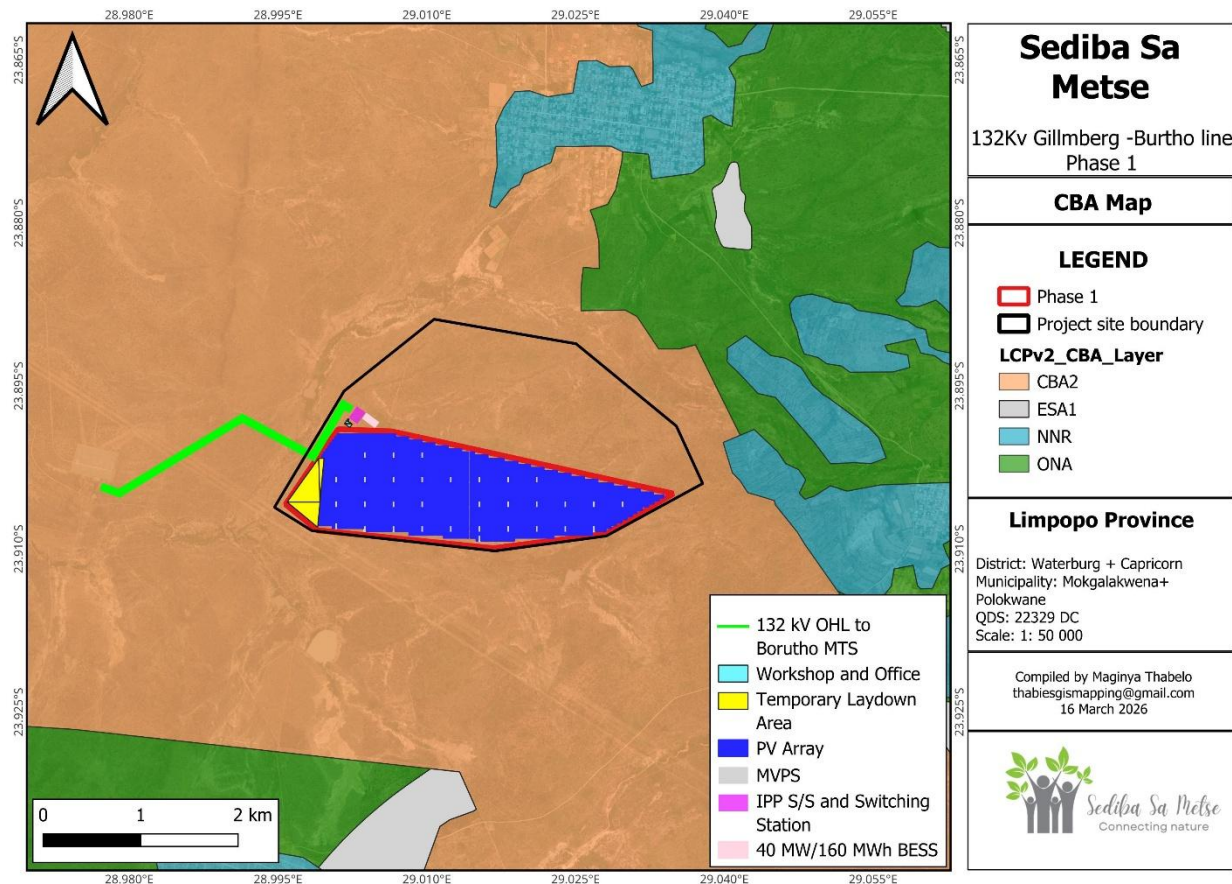


Figure 3: Conservation map for the Phase 1 Gilimberg Solar PV Farm.

13 KEY BIODIVERSITY AREAS

Site-based management efforts are among the most effective means to reduce global biodiversity loss. Therefore, it is crucial to identify sites with globally significant biodiversity, so that we can prioritise where and how to act most effectively. Key Biodiversity Areas are the most important places in the world for species and their habitats. Through the Key Biodiversity Area (KBA) programme in South Africa, we identify, map, monitor, and conserve these high-priority areas to help safeguard the most critical sites for nature.

SANBI and BirdLife South Africa lead this important initiative in South Africa in collaboration with several other partners.

What are Key Biodiversity Areas, and why are they important?

Key Biodiversity Areas (KBAs) are 'sites that contribute significantly to the global persistence of biodiversity', which means they are the most important places in the world for species and their habitats, whether these be in terrestrial, freshwater, estuarine, or marine ecosystems. The aim is to help prevent the rapid loss of biodiversity by supporting nationally led efforts to identify places

worldwide that are critical for the survival of unique plants and animals and the ecological communities they comprise. KBAs provide a global currency for identifying the most biologically significant sites, thereby focusing our collective efforts on conserving the places that matter most. Sites qualify as global KBAs if they meet the specific standardised criteria and quantitative thresholds focused on one or more of the five trigger aspects:

Threatened biodiversity	Sites that support significant populations of species or ecosystems at risk of extinction.
Geographically restricted biodiversity	Sites that hold species or ecosystems with limited global ranges or extent.
Ecological integrity	Sites with intact ecological communities that maintain natural processes and have low human impact.
Biological processes	Locations critical for key life-cycle events such as breeding aggregations, refuges, or recruitment sources.
Irreplaceability	Sites essential for achieving conservation goals, where loss would significantly reduce options for biodiversity protection.

According to the South African KBAs (2024), the project area does not fall within any KBA.

14 HYDROLOGY AND AQUATIC

Multiple ephemeral drainage lines are mapped during the study, which form part of the catchment feeding into the broader river systems outside the development footprint (

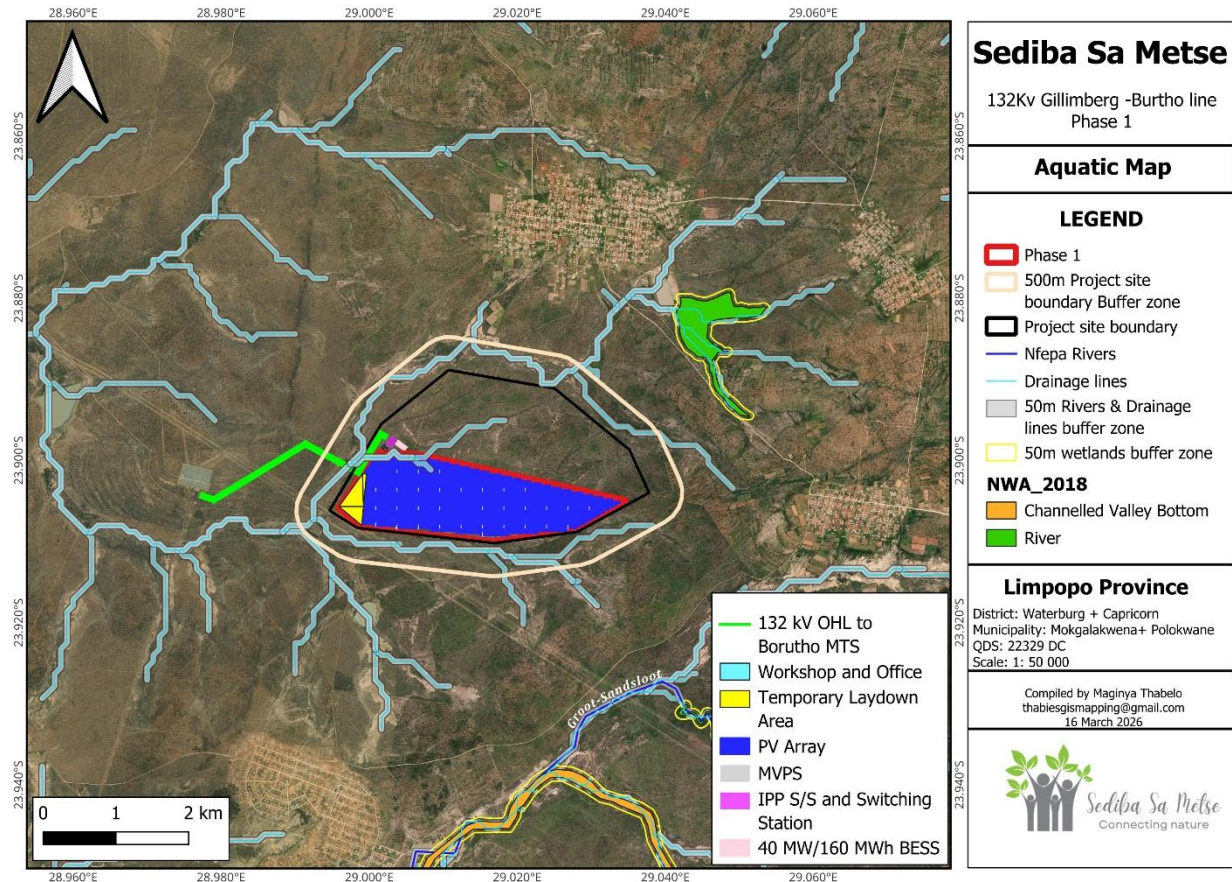


Figure 4). And no permanent wetlands are identified within the 500m regulated zone of the site.

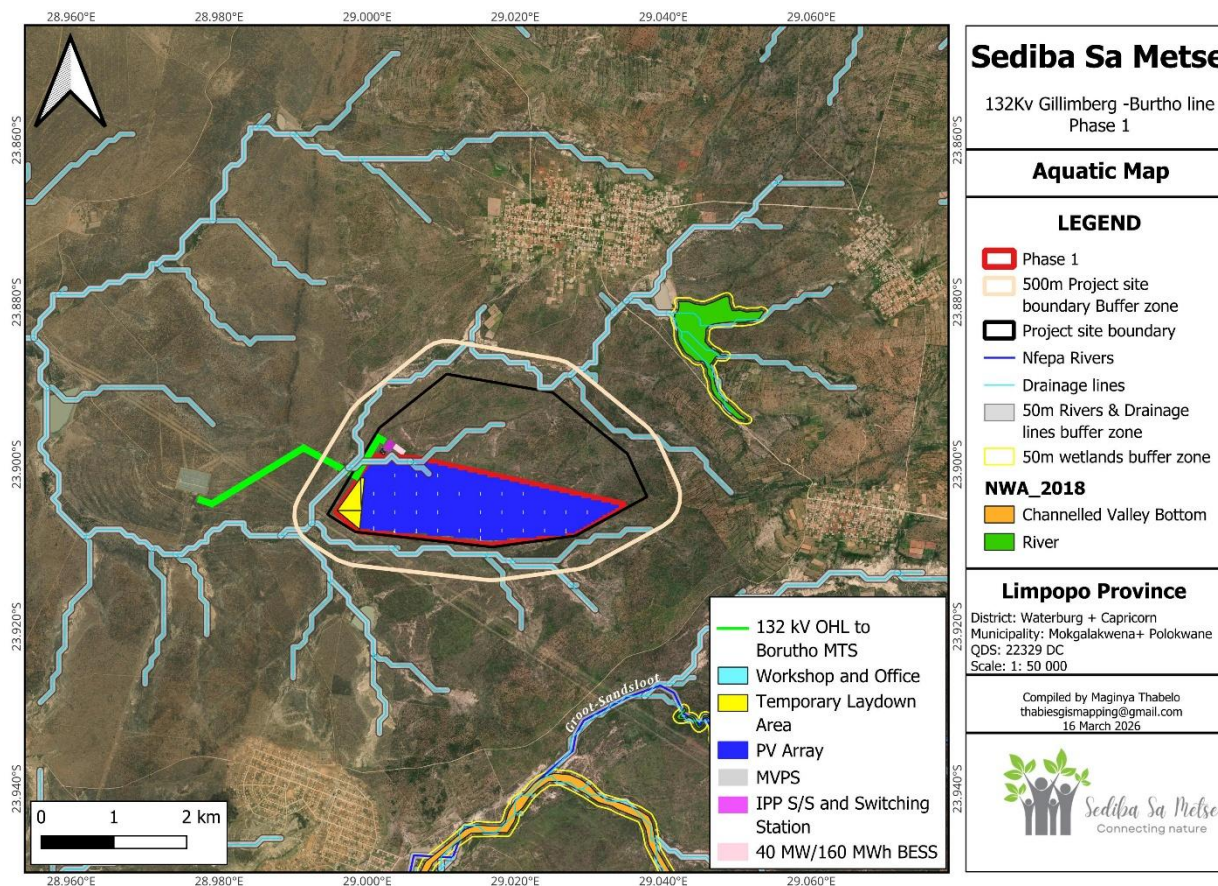


Figure 4: Aquatic features for the Phase 1 Gilimberg Solar Farm.

15 LANDSCAPE AND LOCAL CORRIDORS

For certain movement-related ecological processes to occur, such as species migration, seasonal and altitudinal dispersal, and range displacement in response to climate change, connections must be maintained. Therefore, these corridors primarily aim to maintain regional connectivity to support ecological processes rather than to meet specific biodiversity targets. It should be noted that none of the development components fall within a Landscape Corridor.

16 SOUTH AFRICAN PROTECTED AREAS (2024)

The Protected Areas (2024) dataset was used to determine whether the proposed Phase 1 Gilimberg RE 1, 200MW Cluster Solar PV Facility falls within or near any formally declared protected areas (SANBI, 2024). This dataset represents the most recent spatial inventory of legally declared protected areas in South Africa, including national parks, provincial nature reserves, and private nature reserves recognised under the National Environmental Management: Protected Areas Act (Act 57 of 2003) (Republic of South Africa, 2003; SANBI, 2024). Protected areas are formally designated to conserve biodiversity, maintain ecological

processes, and safeguard representative ecosystems and species habitats, thereby contributing to national and international biodiversity conservation commitments (DEA, 2016; SANBI, 2019). According to the Protected Areas (2024) dataset, the study area does not fall within a protected area or the National Protected Area Expansion Strategy.

17 ECOSYSTEM THREAT STATUS

The NBA primarily focuses on evaluating biodiversity at the ecosystem and species levels in the four realms (terrestrial, inland aquatic, estuarine, and marine), as well as in two cross-realm areas (the coast and South Africa's sub-Antarctic territory), which are subject to the two headline indicators of threat status and protection level (Driver et al., 2011).

Ecosystem Threat Status and Protection Level

The National Environmental Management: Biodiversity Act (Act 10 of 2004) provides for listing threatened or protected ecosystems in one of four categories, namely Critically Endangered (CR), Endangered (EN), Vulnerable (VU), or Least Threatened (LT). The main purpose of listing Threatened Ecosystems is to reduce the rate of ecosystem and species extinctions and to prevent further degradation and loss of structure, function, and composition of threatened ecosystems (Nel & Driver, 2014). Ecosystem threat status outlines the degree to which ecosystems are still intact or, alternatively, are losing vital aspects of their structure, function, and composition, on which their ability to provide ecosystem services ultimately depends, for each ecosystem type that remains in good ecological condition (Driver et al., 2011).

The level of ecological protection indicates whether or not ecosystems are sufficiently protected. Based on the percentage of each ecosystem type within a protected area recognized under the Protected Areas Act, ecosystem types are categorized as not protected, poorly protected, moderately protected, or well protected (Driver et al., 2012).

The National Environmental Management: Biodiversity Act, (Act No. 10 OF 2004) (NEM:BA) provides a National List of Ecosystems that are threatened and in need of protection – GN 1002 of 2011. According to the NEM:BA List of threatened ecosystems (2011), the project area does not fall within any threatened ecosystem ([Figure 5](#)).

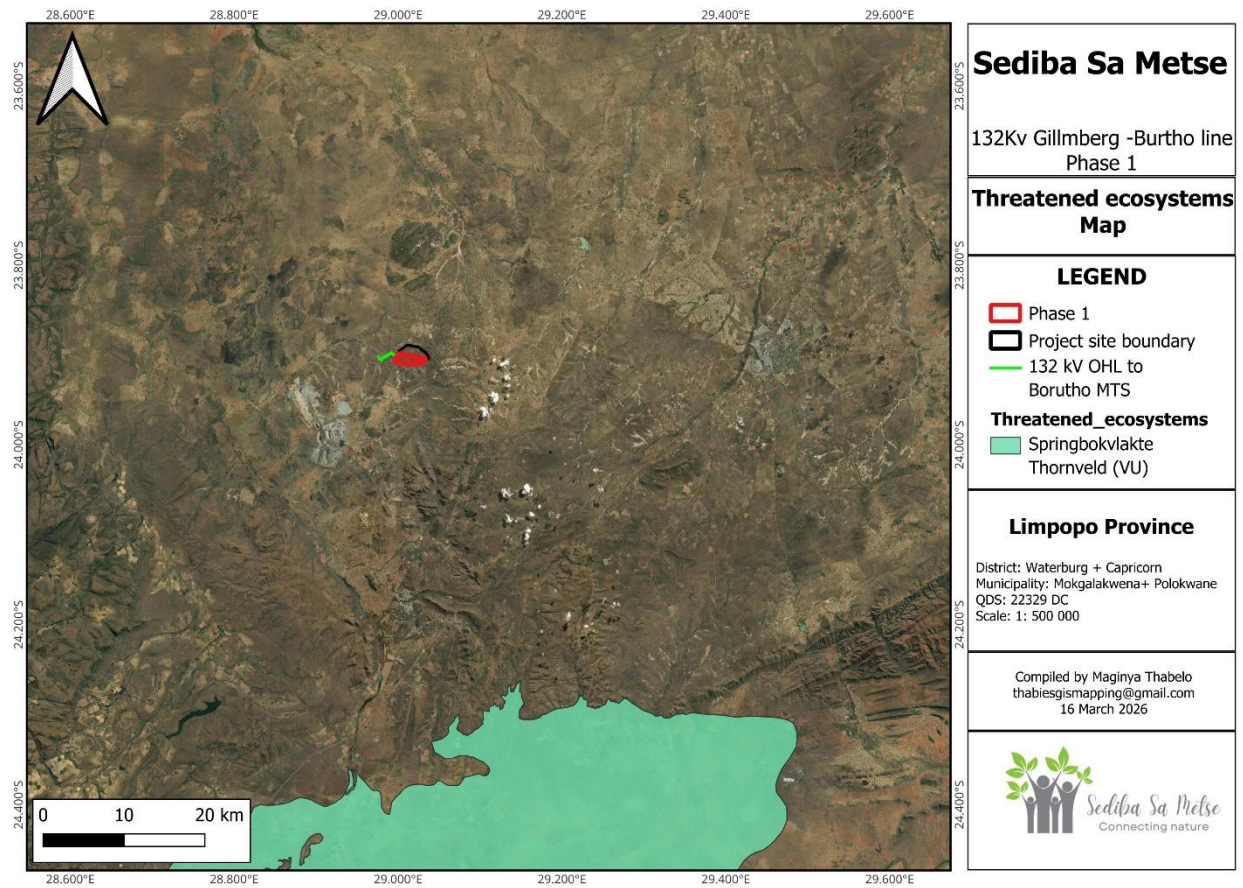


Figure 5: Threatened Ecosystem map for the Phase 1 Gilimberg Solar PV Farm.

18 METHODOLOGY AND ASSESSMENT APPROACH

18.1 Desktop Assessment

A desktop assessment was undertaken using freely available geographic information and mapping tools, including QGIS (v. 3.44.0) and Google Earth Pro (v. 7.3.6). These tools are reliable, accurate, and effective for detecting and analysing present and historical imagery across most of the world, including Limpopo Province. The desktop study consulted available spatial and ecological datasets relevant to the project site and surrounding areas to establish the most recent and reliable information on the ecosystem.

The following activities were undertaken:

- The project footprint was mapped in QGIS and the BGIS to obtain all relevant shapefiles.
- The National Biodiversity Assessment (NBA) datasets on rivers, wetlands, South African vegetation types, Important Biodiversity Areas (IBAs), and Protected areas were overlaid to determine their occurrence within or near the project site.
- Additional SANBI datasets consulted included:
 - **Red List of South African Plants** (Threatened Species Programme) to identify any red-listed plants in the area (<http://redlist.sanbi.org/index.php>, accessed 01 and 20 April 2026).
 - **Plants of South Africa (POSA)** database to identify provincially protected and endemic plant species (<https://posa.sanbi.org/sanbi/Explore>, accessed 01 and 20 April 2026).

A literature review was conducted on publications relevant to the Limpopo Province and to the vegetation type of the study area. This ensured a robust understanding of the ecological context and the most appropriate biodiversity assessment guidelines, processes, and protocols to apply.

Key references included:

- Guidelines for Incorporating Biodiversity into Environmental Impact Assessment (Brownlie et al., 2009).
- Guidelines for Working in Freshwater Ecosystem Priority Areas (Driver et al., 2011).
- National Biodiversity Assessment: Synthesis Report 2018 and Provincial Summaries 2017.
- National Terrestrial Threat Status and Protection Level (2018).

A complete list of all materials and datasets consulted during this assessment is provided in the References section of this report.

18.2 Field Assessment

The specialists conducted a five-day on-site survey. A reconnaissance ‘walkabout’ and point survey (14 plots) was undertaken to determine the general habitat types found throughout the study area (Figure 6). The focus was on identifying species of conservation concern (SCC), Threatened or Protected Species (TOPS), or Red Data Listed (RDL) Species. Our survey primarily targeted ridges, rocky areas, streams, drainage lines, less-disturbed sites, and servitudes.



Figure 6: Vintage Points, Walkabout and drive transect.

18.2.1 Flora

The plant species (trees, succulents, graminoids, geophytes, and forbs) within the site were identified. Species recorded include all possible species that could be recorded at the time of the survey, including all SSC and RDL species, as well as invasive alien plants (IAP). Collection of such data continued throughout the assessment by walking the site at random and plots to ensure a representative sample of the entire study site.

The probability of occurrence (POC) of the species of concern was also taken into consideration. The POC was determined from literature data and habitat status (suitability and disturbance) during the site assessment and expressed as a percentage using the following formula.

$$[\text{Literature Data} + \text{Habitat Suitability} + \text{Habitat disturbance}] / 15 \times 100 = \text{POC}\%$$

The following score in Table 2 was used to rank or rate the above factors, and each contributed equally to the formula. The accuracy of the calculation depends on the available data and specialist knowledge of the species in question.

Table 2: The score and factors used to determine POC.

	Literature Data (from scientific sources)					
Score	0	1	2	3	4	5
	Habitat Suitability (<i>specialist assessment on site</i>)					
Score	0	1	2	3	4	5
	Habitat Disturbance (<i>specialist assessment on site</i>)					
Score	0	1	2	3	4	5

18.2.2 Fauna

For all animal species, it is acknowledged that some may have been missed due to the assessment period, the season, and the limitations of this type of assessment. It must be noted that some species, especially reptiles, birds, and other mammals such as porcupines, are secretive and/or nocturnal. Therefore, the POC prediction model will also apply. Significant consideration was given to microhabitat types and their potential occurrence, as well as to available databases obtained from various sources. The specialists made a great effort to assess this aspect.

18.2.2.1 Mammals

Potential mammal species for the study area were identified through desktop analysis using:

- iNaturalist South Africa (<https://www.inaturalist.org/places/south-africa>)
- The Mammals of the Southern African Subregion (Skinner & Chimimba, 2005), The Red Data Book of the Mammals of South Africa: A Conservation Assessment [Friedmann and Daly (editors), 2004]

Field verification will include observations within and adjacent to the site, recording birds seen or heard, as well as indirect evidence such as feathers, nests, and droppings.

18.2.2.2 Reptiles and Amphibians

The potential presence of reptiles and amphibians was determined primarily through desktop research using:

- iNaturalist South Africa (<https://www.inaturalist.org/places/south-africa>)
- Literature review

Fieldwork was focused on habitat assessments to determine the site's suitability for these taxa. Due to the seasonal nature and often cryptic habits of many species, POC estimates will be mainly informed by habitat condition and locality data rather than direct observations.

18.2.2.3 Invertebrates

The potential presence of invertebrates was determined primarily through desktop research using:

- iNaturalist South Africa (<https://www.inaturalist.org/places/south-africa>)
- Literature review

19 RED DATA LISTED SPECIES

A desktop assessment was also conducted to screen key, threatened, or species of concern within the study area. The aim was to obtain and provide a list of potentially threatened species that may occur, as highlighted, and to verify the site assessment (see Appendix I).

The primary resources used during the desktop assessment include the National Red List of Threatened Plants of South Africa, Lesotho & Swaziland (SANBI) and the POSA of SANBI, the SABAP2 dataset, the African Snakebite Institute Snakes Application (ASI Snakes), and iNaturalist.

The presence of species of concern, rare and protected species, or suitable habitat was searched for and recorded during the field assessment. The definitions of status and category for the Red Data Listed Species are provided in Table 3 below.

Table 3: Red List Data Status and definitions as per SANBI (2010) guidelines.

P- Protected Species		M- Medicinal species
EX	Extinct	A taxon is Extinct when there is no reasonable doubt that the last individual has died. Taxa should be listed as extinct only once exhaustive surveys throughout the historic range have failed to record an individual.
EW	Extinct in the Wild	A taxon is Extinct in the Wild when it is known to survive only in cultivation or as a naturalized population (or populations) well outside the past range.
CR PE	Critically Endangered (Possibly Extinct)	Critically Endangered (Possibly Extinct) taxa are those that are, on the balance of evidence, likely to be extinct, but for which there is a small chance that they may be extant. Hence, they should not be listed as Extinct until adequate surveys have failed to record the taxon.
CR	Critically Endangered	A taxon is Critically Endangered when the best available evidence indicates that it meets any of the five IUCN criteria for Critically Endangered and is therefore facing an extremely high risk of extinction in the wild.
EN	Endangered	A taxon is Endangered when the best available evidence indicates that it meets any of the five IUCN criteria for Endangered and is therefore facing a very high risk of extinction in the wild.
VU	Vulnerable	A taxon is Vulnerable when the best available evidence indicates that it meets any of the five IUCN criteria for Vulnerable and is therefore facing a high risk of extinction in the wild.
NT	Near Threatened	A taxon is Near Threatened when available evidence indicates that it nearly meets any of the five IUCN criteria for Vulnerable and is therefore likely to qualify for a threatened category in the near future.
CRITICALLY RARE		A taxon is Critically Rare when it is known to occur only at a single site but is not exposed to any direct or plausible potential threat and does not qualify for a category of threat according to the five IUCN criteria.

P- Protected Species		M- Medicinal species
RARE		A taxon is Rare when it meets any of the four South African criteria for rarity but is not exposed to any direct or plausible potential threat and does not qualify for a category of threat according to the five IUCN criteria.
DECLINING		A taxon is Declining when it does not meet any of the five IUCN criteria and does not qualify for the categories Critically Endangered, Endangered, Vulnerable or Near Threatened, but there are threatening processes causing a continuing decline in the population.
DDD	Data Deficient— Insufficient Information	A taxon is DDD when there is inadequate information to make an assessment of its risk of extinction, but the taxon is well defined. Data Deficient is not a category of threat. However, listing of taxa in this category indicates that more information is required, and that future research could show that a threatened classification is appropriate.
LC	Least Concern	A taxon is Least Concern when it has been evaluated against the five IUCN criteria and does not qualify for the categories Critically Endangered, Endangered, Vulnerable or Near Threatened, and it is not rare, and the population is not declining.

20 DATA ANALYSIS

Any large-scale development within a natural system inevitably exerts pressure on the environment, often with negative ecological consequences if not carefully managed. It is therefore critical during the planning and environmental authorisation phase to identify potential impacts and design effective mitigation measures. Best practice requires that such developments be located in already modified or fragmented areas wherever possible (Edwards et al., 2018).

20.1 Assumptions and Limitations

The findings of this assessment are subject to the following assumptions and limitations:

- It is assumed that all third-party data sources (e.g., SANBI BGIS, Limpopo Conservation Plan shapefiles, Google Earth imagery) used in this assessment are accurate and up to date;
- Evidence of fauna was determined through both direct observations and indirect signs (e.g., tracks, scats, nests, burrows, and shed skins). Some species may remain undetected due to cryptic behaviour, low densities, or nocturnal activity patterns;
- It is acknowledged that additional biodiversity records may become available after completion of this report through ongoing databases such as SANBI and citizen science platforms (e.g., iNaturalist);
- The project site is located within a semi-natural landscape surrounded by villages, agricultural activities, and grazing areas, which may influence species composition and distribution;
- The survey was limited to a single 4-5-day field assessment, which may restrict the detection of some species and reduce overall sampling completeness;
- Survey conditions were predominantly cloudy with intermittent rainfall, which is not optimal for biodiversity surveys and likely influenced detectability. These conditions may have resulted in:
 - Reduced activity of reptiles and many invertebrates, which are less active under cool and wet conditions;
 - Reduced avian activity and vocalisation, limiting detection of bird species;
 - Variable detectability across taxa due to the combined effects of temperature, cloud cover, and rainfall;
- Seasonal constraints: The site visit was conducted in March 2026, representing a specific seasonal window. As such:

- Some flora species may not have been identifiable outside flowering or fruiting periods;
- Certain faunal species (e.g., migratory birds, amphibians, invertebrates) may exhibit seasonal or weather-dependent activity patterns and may have been absent or less detectable;
- Although rainfall can increase activity for some taxa (e.g., amphibians), the prevailing conditions may have reduced overall detectability across multiple groups;
- Invertebrate assessment was conducted at a habitat level, and no detailed taxonomic sampling, trapping, or specialist collection methods were undertaken;
- The likelihood of occurrence for Species of Conservation Concern (SCC) is therefore based on habitat suitability, known distribution ranges, and specialist judgment, rather than direct observation alone;
- Additional ecological information may become available after completion of this assessment, which may influence interpretations and conclusions.

Despite these limitations, the integration of desktop analysis, field observations, and specialist expertise provides a robust and reliable assessment of the ecological sensitivity, biodiversity value, and potential impacts associated with the proposed development.

21 RESULTS

21.1 Vegetation

The overall vegetation of the project site, Figure 7.



Figure 7: The overall vegetation of the site

A total of **95 identified plant species** were recorded within the project area (Table 3). Over 92% is dominated by indigenous species. The vegetation at the time of the survey was semi-natural. There are two protected species on site. Protected species, Sensitive Plant 1252 and the Marula tree (*Sclerocarya birrea afra*) (Figure 8) have been observed on site.



Marula tree observed on site. The population was abundant and consist of different age classes.

Figure 8: The Marula Tree

Table 4: List of flora found on site.

NO.	Scientific name	Common name	SANBI RED LIST	PROTECTED SPECIES
1	<i>Abutilon austroafricanum</i>		LC	
2	<i>Achyranthes aspera</i>	Devil's Horsewhip	Exotic	
3	<i>Acokanthera oppositifolia</i>	bushmans poison	LC	
4	<i>Acokanthera rotundata</i>	Round-leaf Poison-bush	LC	
5	<i>Acrotome inflata</i>	Horsefright	LC	
6	<i>Aloe greatheadii</i>	spotted aloe	LC	
7	<i>Aloe marlothii</i>	mountain aloe	LC	
8	<i>Antizoma angustifolia</i>		LC	
9	<i>Apodytes dimidiata</i>	White Pear	LC	
10	<i>Aptosimum lineare</i>	Bushveld Carpet Flower	LC	
11	<i>Asparagus virgatus</i>	African Broom Asparagus	LC	

12	<i>Blepharis subvolubilis</i>	Eye Lashes	LC	
13	<i>Bolboschoenus glaucus</i>	tuberous bulrush	LC	
14	<i>Buddleja saligna</i>	False Olive	LC	
15	<i>Carissa bispinosa</i>	num-num	LC	
16	<i>Carissa spinarum</i>	Climbing Numnum	LC	
17	<i>Cleome maculata</i>	Spotted Spindlepod	LC	
18	<i>Cleome monophylla</i>	Singleleaf Spindlepod	LC	
19	<i>Cleome rubella</i>	Pretty Lady	LC	
20	<i>Combretum molle</i>	Fluweelboswilg	LC	
21	<i>Cucumis metuliferus</i>	Jelly Melon / Rooikomkommer	LC	
22	<i>Datura ferox</i>	White Stinkweed	1b	
23	<i>Dicerocaryum senecioides</i>	Savanna Devilthorn	LC	
24	<i>Dichrostachys cinerea</i>	sicklebush	LC	
25	<i>Diospyros lycioides</i>	Quilted Bluebush	LC	
26	<i>Dombeya rotundifolia</i>	African Wild Pear	LC	
27	<i>Drimia altissima</i>	Tall Squill	LC	
28	<i>Ehretia rigida</i>	Puzzle Bush	LC	
29	<i>Eragrostis superba</i>	Heartseed Lovegrass	LC	
30	<i>Euclea crispa</i>	Blue Guarri	LC	
31	<i>Euclea undulata</i>	Common Guarri	LC	
32	<i>Euphorbia cooperi</i>	Candelabra Naboom	LC	
33	<i>Euphorbia ingens</i>	Common Naboom	LC	
34	<i>Euphorbia tirucalli</i>	Rubber Milkbush	LC	
35	<i>Ficus burkei</i>	Common Wild Fig	LC	
36	<i>Flaveria bidentis</i>	speedyweed	1b	
37	<i>Gomphocarpus tomentosus</i> <i>tomentosus</i>	Common Woolly Milkweed	LC	
38	<i>Gossypium herbaceum</i> <i>africanum</i>	Wild Cotton	LC	
39	<i>Grewia flavescens</i>	Sandpaper Raisin	LC	
40	<i>Grewia retinervis</i>	Kalahari Raisin	LC	
41	<i>Gymnosporia buxifolia</i>	Common Spikethorn	LC	
42	<i>Gymnosporia senegalensis</i>	confetti spikethorn	LC	
43	<i>Heteromorpha arborescens</i>	mMakatlala	LC	

44	<i>Hypoestes forskoolii</i>	White Ribbon Flower	LC	
45	<i>Kalanchoe paniculata</i>		LC	
46	<i>Kalanchoe sexangularis</i>	Bushveld Kalanchoe	LC	
47	<i>Kleinia longiflora</i>	paintbrush flower	LC	
48	<i>Ledebouria luteola</i>	highveld African hyacinth	LC	
49	<i>Leucas capensis</i>	White Lionspaw	LC	
50	<i>Leucas martinicensis</i>	White Tumbleweed	LC	
51	<i>Lippia javanica</i>	Fever Tea	LC	
52	<i>Lopholaena coriifolia</i>	Small-leaved Fluff-bush	LC	
53	<i>Maerua afra</i>	Bush Maerua	LC	
54	<i>Melhanian prostrata</i>	Common Melhanian	LC	
55	<i>Melia azedarach</i>	Syringa	1b	
56	<i>Mystroxydon aethiopicum</i>	Kooboo-berry	LC	
57	<i>Olea europaea africana</i>	African Wild Olive	LC	
58	<i>Opuntia ficus-indica</i>	Prickly Pear Cactus	1b	
59	<i>Osteospermum muricatum</i>	Buchu Boneseed	LC	
60	<i>Pappea capensis</i>	Jacket plum	LC	
61	<i>Pellaea calomelanos</i> <i>calomelanos</i>		LC	
62	<i>Peltophorum africanum</i>	Weeping wattle	LC	
63	<i>Persicaria decipiens</i>	slender knotweed	LC	
64	<i>Philyrophyllum schinzii</i>		LC	
65	<i>Plumbago auriculata</i>	blue plumbago	LC	
66	<i>Pogonarthria squarrosa</i>	Herringbone Grass	LC	
67	<i>Pollichia campestris</i>	Waxberry Plant	LC	
68	<i>Salvia leucantha</i>	Mexican Bush Sage	LC	
69	<i>Sansevieria aethiopica</i>	Mother-in-law's Tongue	LC	
70	<i>Sansevieria trifasciata</i>	Snake Plant	LC	
71	<i>Schinus molle</i>	Peruvian Pepper Tree	Invasive (not listed)	
72	<i>Schotia brachypetala</i>	Weeping Boerbean	LC	
73	<i>Sclerocarya birrea afra</i>	Marula	LC	Protected Tree

74	<i>Searsia lancea</i>	Karee	LC	
75	<i>Searsia leptodictya</i>	Mountain Karree	LC	
76	<i>Searsia pyroides</i>	Common Wild Currant	LC	
77	<i>Searsia tumulicola</i>	Hardleaf Curranthrus	LC	
78	<i>Senecio barbertonicus</i>	Succulent Bush Ragwort	LC	
79	<i>Senecio pleistocephalus</i>		LC	
80	<i>Senegalia afra</i>	Common Hook-thorn Acacia	LC	
81	<i>Solanum campylacanthum</i>	bitter-apple	LC	
82	<i>Tagetes minuta</i>	Tall Khaki Weed	LC	
83	<i>Themeda triandra</i>	Red Grass	LC	
84	<i>Vachellia hebeclada</i>	candlepod thorn	LC	
85	<i>Vachellia karroo</i>	Sweetthorn	LC	
86	<i>Vachellia nilotica</i>	Egyptian acacia	LC	
87	<i>Vachellia robusta</i>	moOka	LC	
88	<i>Vachellia tortilis</i>	umbrella thorn	LC	
89	<i>Vangueria infausta</i>	wild-medlar	LC	
90	<i>Vangueria parvifolia</i>	Bergwildemispel	LC	
91	<i>Volkameria glabra</i>	Cat's-whiskers	LC	
92	<i>Xanthium strumarium</i>	rough cocklebur	1b	
93	<i>Xenostegia tridentata</i>	Narrowleaf Merremia	LC	
94	<i>Zantedeschia albomaculata</i>	Spotted Arum	LC	
95	<i>Ziziphus mucronata</i>	buffalo-thorn	LC	

21.2 Plant Species of Conservation Concern

The site lies within the least concern Polokwane Plateau Bushveld and the vulnerable Makhado Sweet Bushveld, and overlaps CBA2, with the DFFE screening indicating possible sensitive taxa in the area (*Sensitive species 1252*). These flags were considered alongside field evidence and local habitat suitability when confirming SCC on site.

According to the DFFE Screening Tool Report, the plant species theme for the project area has been classified as "Medium" sensitivity, with one Sensitive Plant Species (*Sensitive species 1252*) potentially occurring on site. This assessment is based on species distribution modeling, which incorporates known habitat preferences together with the proximity of verified occurrence records. The field survey confirmed that the species has a low likelihood of

occurrence (see Table 4). In addition, one expected SCC (*Elaeodendron transvaalense*), identified based on a desktop assessment, has a high likelihood of occurrence.

To determine the presence of Species of Conservation Concern (SCC) within the proposed development footprint, information was collated from the DFFE Screening Tool Report, iNaturalist records, the Plants of Southern Africa (POSA) database, and relevant regional studies. Each species was assessed for its likelihood of occurrence based on the availability of suitable habitat within the site, the proximity of previously recorded populations, and whether the project area falls within the species' known distribution.

Table 5: Plant SCC with a distribution that includes the project area and the likelihood of occurrence within the project area.

*The Species Environmental Assessment Guideline (SANBI, 2020) specifies the likelihood of occurrence as Low, Moderate, and High. For this assessment, Low=Unlikely to occur, Moderate=Possible occurrence, and High = Probable occurrence.

No.	Species Name	IUCN Red List	Habitat preference	Flowering time	Probability of Occurrence
1	Sensitive 1252	VU	It grows predominantly in moister bushveld areas, forest margin, bracken or ericoid scrub, Albany thicket, fynbos, grassland, savanna, coastal bush and wooded mountain kloofs. It occurs from the coast to the mountains, at an altitude ranging from 125 to 1 650 m above sea level.	summer (November to April)	Low
2	<i>Elaeodendron transvaalense</i>	NT	Savanna or bushveld, from open woodland to thickets, often on termite mounds.	summer (December to April)	High

21.3 Alien Invasive Plant Species

Alien Invasive Plant (AIP) species are plants that occur outside their natural geographic range, having been introduced by human activities, either intentionally or accidentally. While not all alien species are invasive, and not all invasive species are alien, specific taxa can establish and spread aggressively, leading to ecological disruption. These species often displace native

vegetation, alter ecosystem processes, and reduce habitat quality for indigenous flora and fauna.

In terms of the National Environmental Management: Biodiversity Act (NEM:BA, Act No. 10 of 2004), an Invasive Alien Plant Species is defined as any taxon whose establishment and spread outside its natural distribution range:

- (a) threatens ecosystems, habitats, or other species, or has the potential to do so; and
- (b) may result in economic, ecological, or human health impacts.

During the field survey, seven exotic species were recorded scattered within the proposed development footprint, of which six are invasive (see [Table 3](#)). These species are listed under the NEM:BA National List of Invasive Species (2004; amended 2020). In accordance with the legislation, all Category 1b invasive plants must be controlled or eradicated, and their further spread is strictly prohibited. No propagation, trade, or planting of such species is permitted within South Africa, and category two species are allowed only in specified areas under controlled conditions.

22 ANIMAL SPECIES THEME

22.1 Fauna species distribution in relation to the project area

There are over 73 bird species recorded from the project area, including the Project Area of Influence (PAOI). There is an observation of the Critically Endangered species, the White-backed Vulture, Figure 9.



Figure 9: White-backed Vulture.

A total of **seven identified faunal species** were recorded: three mammals and four reptiles (Table 5). This data excludes those that could not be identified using tracks and scats. No SCC was observed during the survey.

Table 6: Faunal list observed on site.

No.	Scientific name	Common name	IUCN status	Protected species
MAMMALS				
1	<i>Herpestes sanguineus</i>	Common Slender Mongoose	LC	
2	<i>Paraxerus cepapi</i>	Smith's Bush Squirrel	LC	
3	<i>Sylvicapra grimmia</i>	Common Duiker	LC	
REPTILES				
1	<i>Acanthocercus atricollis</i>	Southern Tree Agama	LC	LEMA protected
2	<i>Agama aculeata</i>	Ground Agama	LC	LEMA protected
3	<i>Trachylepis varia</i>	Variable Skink	LC	LEMA protected
4	<i>Trachylepis punctatissima</i>	Speckled Rock Skink	LC	LEMA protected

22.2 Cultural and Socio-Economic Importance of Invertebrates

Statement that invertebrates were assessed at the habitat level, not species-specific, due to survey limitations. The present study reported only common invertebrates. The invertebrate assemblage expected within the project area includes species that are not only ecologically important but also hold cultural, traditional, and socio-economic value within local communities.

Of particular relevance is the presence or potential occurrence of edible and culturally significant insect species, such as the Giant Jewel Beetle.

22.3 Faunal Species of Conservation Concern

According to the **DFFE Screening Tool Report for Animal Species**, the project area has been classified as “**Medium sensitivity**” for fauna, primarily due to the medium potential presence of Mammalia-*Dasymys robertsii*. The species has a low likelihood of occurrence within the project area (Table 6) due to the persisting current impacts (livestock, firewood collection, and Present Ecological Status (PES). Based on desktop and field observations, no additional SCC is expected.

Table 7: Faunal SCC with a distribution that includes the project area and the likelihood of occurrence within the project area.

*The Species Environmental Assessment Guideline (SANBI, 2020) specifies the likelihood of occurrence as Low, Moderate, and High. For this assessment, Low=Unlikely to occur, Moderate=Possible occurrence, and High = Probable occurrence.

No.	Scientific Name	Common Name	Conservation Status	Habitat preference	Distribution includes or partly includes the project area	Preferred habitat available in the project area	Likelihood of Occurrence	Justification
1	<i>Dasymys robertsii</i>	Robert's Shaggy Rat	VU	These species are associated with rivers and wetlands. These species have been recorded from a wide variety of habitats, including forests and savannas, swamplands and grasslands, but they rely on intact wetlands in these areas. They have not been recorded from agricultural landscapes or dam areas. They occur specifically in reed beds and among semi-aquatic grasses in wetlands or swampy areas or along rivers and streams, as well as in grassy areas close to water, wherein they co-occur with <i>Otomys</i> spp. Roberts's Shaggy Rat is primarily herbivorous,	Yes	No	Low	The species is strongly associated with permanent wetlands, including reed beds and dense semi-aquatic vegetation along rivers and streams. Although the broader distribution overlaps the project area, no suitable preferred habitat (i.e., intact wetlands or permanent water bodies) is present on site. The project area is limited to ephemeral drainage lines, which do not provide the consistent moisture, vegetation structure, or ecological

				feeding mostly on aquatic and terrestrial plants, but to a lesser extent also on insects.				conditions required by the species.
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23 CONSERVATION AND SENSITIVE AREAS

The proposed Phase 1 Gilmberg RE 1, 200MW Cluster Solar PV Facility and associated infrastructure, including its 2km overhead powerline, is located in the area that is classified as a medium sensitivity to disturbance due to its diverse indigenous vegetation (semi-natural veld). The area is largely semi-natural, having been historically utilised for game farming and subsistence agriculture, and is currently used for livestock grazing. Additional disturbances include firewood collection and localised sand mining, which have resulted in moderate habitat degradation but not full transformation of the vegetation (Figure 10 and Figure 12 below). The only areas that have been significantly transformed within the broader project context are associated with existing Eskom infrastructure, namely the substation and powerline corridor (see Figure 10).

One protected species was recorded during the site assessment. In addition, there is a high likelihood of occurrence of the Near Threatened and protected species, *E. transvaalense*, particularly within less disturbed vegetation patches.

The Vulnerable Makhado Sweet Bushveld vegetation type occurs along the powerline corridor connecting the solar development sites. Although this vegetation type is of conservation concern, the affected portion has already been disturbed by existing infrastructure (see Figure 2 above).

The site retains ecological functionality, with habitat provision and limited landscape connectivity. However, these functions are moderately reduced due to ongoing anthropogenic pressures.

Overall, the presence of a protected species, the potential occurrence of additional species of conservation concern, and the inclusion of a Vulnerable vegetation type support a moderate conservation sensitivity classification.

23.1 SENSITIVE AREAS

Sensitive areas within the project site are defined as habitats supporting species of conservation concern, remaining semi-natural vegetation, and areas associated with vulnerable vegetation types. Mainly, these areas are the drainage lines, Figure 11. The following sensitivities were identified:

- Species of Conservation Concern (SCC): One protected species was confirmed on-site. There is also a high likelihood of *E. transvaalense* occurring, which constitutes a key ecological constraint, but this survey did not observe this species.
- Vegetation Sensitivity: The Vulnerable Makhado Sweet Bushveld vegetation type is present within the broader project area, particularly along the powerline corridor.

Although partially disturbed, it remains of conservation importance. (see [Figure 2 above](#)).

- **Disturbance Regime:** The site is semi-natural but subject to grazing, firewood harvesting, and localised sand mining, resulting in moderate disturbance levels.
- **Ecological Function:** Remaining vegetation continues to provide habitat and contributes to ecological connectivity, albeit at a reduced level due to fragmentation and disturbance.

Given the semi-natural condition of the site and existing disturbances, the proposed development is anticipated to result in a medium-low impact on the vegetation type, provided that mitigation measures are implemented and sensitive features are appropriately managed and avoided where possible.

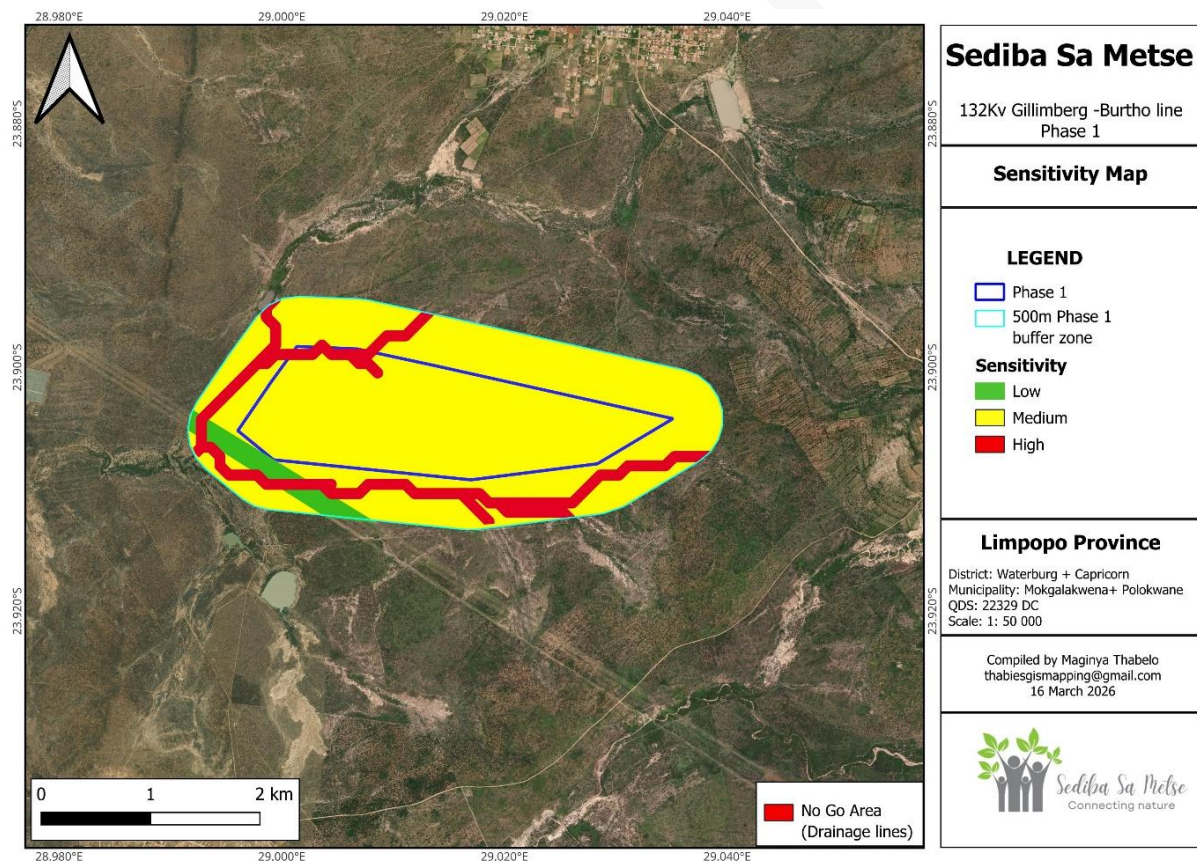


Figure 10: The sensitivity Map for Phase 1 Gilimberg Solar PV Farm.

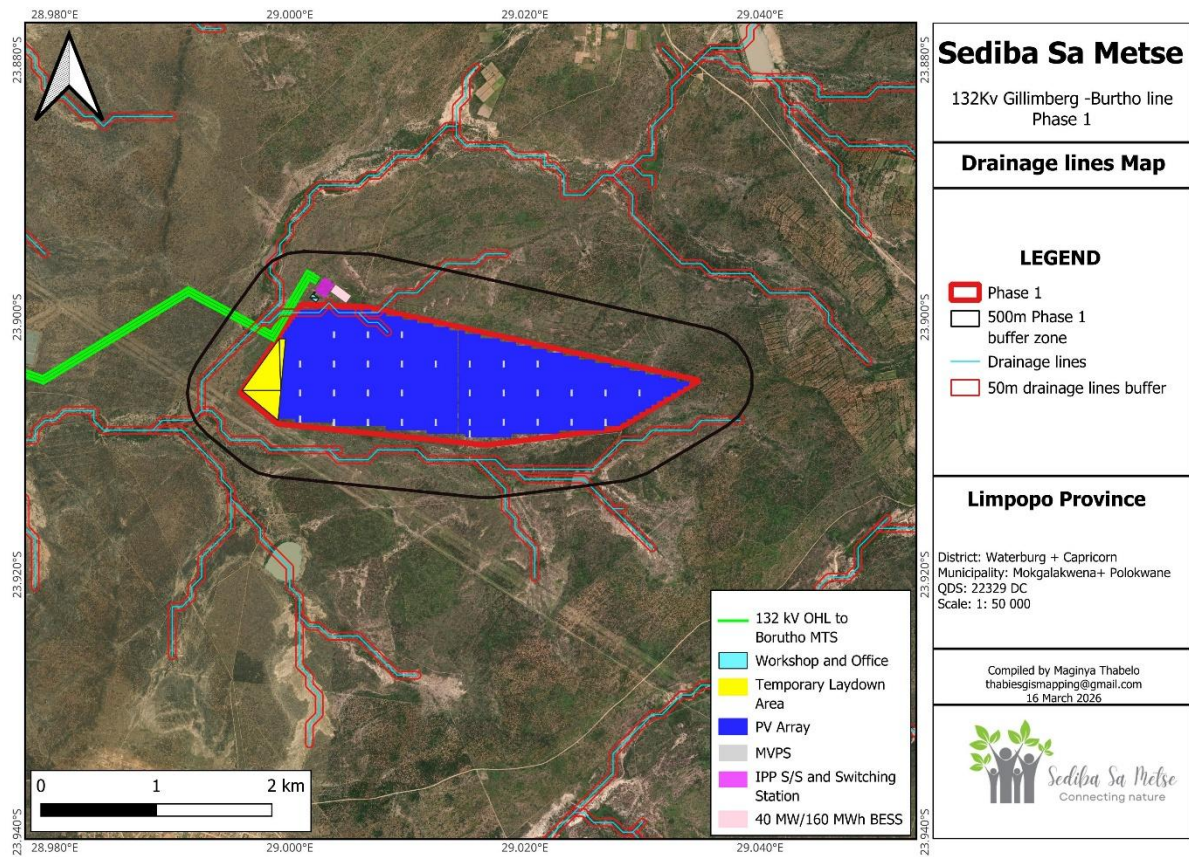


Figure 11: Sensitive Areas (Drainage Lines) of the Phase 1 Gilimberg Solar PV Farm.



Figure 12: The drainage line at the project site.

24 ECOLOGICAL FUNCTION

Ecological function relates to the degree of ecological connectivity between systems within a landscape matrix. Those systems with a high degree of ecological connectivity within the landscape are perceived as more sensitive and are likely to contribute to ecosystem services (e.g., wetlands, ridges) or to the overall preservation of biodiversity. Conservation importance relates to species diversity, endemism (unique species or processes), and the high occurrence of threatened and protected species or ecosystems that are protected by legislation.

Description of the Sensitivity scale

- **No Go Areas:** Areas that have irreplaceable biodiversity or important ecosystem function values which may be lost permanently if these ecosystems are transformed, with a high potential of also affecting adjacent and/or downstream ecosystems negatively.
- **High ecological function:** Sensitive ecosystems with either low inherent resistance or resilience towards disturbance factors or highly dynamic systems considered to be stable and important for the maintenance of ecosystem integrity for example, pristine grasslands, pristine wetlands, and pristine ridges.
- **Medium ecological function:** Relatively important ecosystems at gradients of intermediate disturbances. An area may be considered of medium ecological function if it is directly adjacent to a sensitive/pristine ecosystem.
- **Low ecological function:** Degraded and highly disturbed systems with little or no ecological function.

Conservation status of the vegetation

- **High conservation importance:** Ecosystems with high species richness, which usually provide suitable habitats for several threatened species. Traditionally termed 'no-go' areas, and unsuitable for development and should be conserved.
- **Medium conservation importance:** Ecosystems with intermediate levels of species diversity without any threatened species. Low-density development may be accommodated, provided the current species diversity is conserved.
- **Low conservation importance:** Areas with little or no conservation potential and usually species-poor (most species are generally exotic).

The 300ha project area is considered to have a moderate level of ecological function. The site remains largely semi-natural, supporting ongoing ecological processes and providing habitat for a range of species. The area also contributes to local ecological connectivity, functioning as a movement corridor for fauna and supporting floral continuity within the broader landscape. However, ecological function is moderately reduced due to existing anthropogenic pressures, including livestock grazing, firewood collection, and localised sand mining. In addition, the presence of existing Eskom infrastructure has resulted in some habitat fragmentation, particularly along the powerline corridor.

The site falls within a Vulnerable vegetation type (Makhado Sweet Bushveld), and one protected species was confirmed during the field assessment. Furthermore, there is a high likelihood of *E. transvaalense* occurring, indicating the presence of conservation-relevant ecological features.

Despite these sensitivities, the site is not an intact or pristine ecosystem and is not considered irreplaceable for biodiversity conservation.

This confers a Medium Ecological Function.

25 SITE ECOLOGICAL IMPORTANCE (SEI)

The ecological sensitivity assessment, based on Present Ecological Status (PES) and Ecological Importance and Sensitivity (EIS), aligns with the Site Ecological Importance (SEI) framework described in the Species Environmental Assessment Guideline.

The site demonstrates:

- Moderate biodiversity importance, due to the presence of semi-natural vegetation and SCC;
- Moderate functional integrity, with some disturbance but retained ecological processes;
- Moderate resilience, with the ability to recover under appropriate management;

Accordingly, the site is considered to have medium ecological importance, consistent with the SEI approach.

26 CONSERVATION IMPORTANCE

The site supports **moderate species diversity**, including one confirmed protected species and the potential presence of additional species of conservation concern, *E. transvaalense*. While the vegetation is largely semi-natural, it has been subject to disturbance and does not represent a pristine or highly diverse system.

Given the absence of multiple confirmed threatened species and the moderate level of disturbance, the site is assigned **Medium Conservation Importance**. The area remains suitable for development, provided that biodiversity features are appropriately managed and conserved (e.g., avoid the drainage lines) where possible.

Present Ecological Status (PES)

This section summarizes the system's present ecological status (PES) and gives the Ecological Importance and Sensitivity (EIS) of the site. The status is derived from integrating the desktop and field assessment outcomes. The Ecological Importance (EI), which reflects the importance of the ecosystem in maintaining biological diversity and ecological functioning at the local/regional and landscape levels (Kotze et al., 2020), was determined. Ecological Sensitivity (ES) refers to the ecosystem's fragility or the ability to resist or recover from a disturbance (Kotze et al., 2020). The proposed site is presented as:

Table 8: Present Ecological Status.

Proposed Site	PES	EIS	Recommendation
Phase 1 Gilimberg Solar PV Farm	Medium	Medium	The proposed development is considered environmentally feasible, provided that mitigation measures are implemented. Sensitive features, including confirmed protected species and potential individuals of <i>Elaeodendron transvaalense</i> , must be rescued and relocated , and their survival must be ensured during wet season for speedy recovery. Areas outside the proposed 300ha must be avoided and left undisturbed. Disturbance to the Vulnerable Makhado Sweet Bushveld, particularly along the connecting 2km powerline corridor, should be minimised. All drainage features must be protected with appropriate buffers (50m minimum, the kmz file provided). An alien invasive species control programme, erosion management measures, and progressive rehabilitation must be implemented. Monitoring of species of conservation concern and avifaunal should be undertaken during construction and operation to ensure compliance and minimise impacts.

27 IMPACTS DISCUSSION

Any development within a natural system will exert ecological impacts, typically negative unless managed through well-designed mitigation measures. As emphasised by Edwards *et al.*, (2018), priority should always be given to placing developments in already modified or fragmented areas, thereby reducing pressure on natural ecosystems.

The 300ha proposed site is considered to have a moderate ecological importance and sensitivity (medium EIS). This is based on the semi-natural condition of the vegetation, the presence of one confirmed protected species, and the high likelihood of *E. transvaalense* occurring. While the site retains ecological value, it is not considered pristine or irreplaceable, and impacts can be effectively managed through appropriate mitigation.

The potential impacts are discussed by theme below.

Process Impacts – Connectivity and Edge Disturbance

The semi-natural vegetation within the project area contributes to local ecological connectivity, allowing the movement of fauna and supporting the continuity of plant communities. However, this connectivity is already moderately affected by existing land uses and infrastructure.

Key impacts include:

- Localised loss of habitat and foraging areas for birds, small mammals, and invertebrates.
- Partial disruption of ecological connectivity within the immediate project footprint.
- Edge effects resulting in changes to microclimate (e.g., increased light, wind exposure, and temperature fluctuations), which may affect sensitive plant species and associated fauna.

Given the existing disturbance regime, these impacts are expected to be **moderate and localised**.

Pollution Impacts – Soil, Air, and Noise

Construction and operational activities will generate pollution pressures that may affect ecological receptors.

Key impacts include:

- Noise disturbance to fauna, particularly birds and small mammals, potentially affecting behaviour and breeding patterns.
- Dust generation from construction activities and vehicle movement, which may reduce photosynthetic efficiency in nearby vegetation and affect habitat quality.
- Solid waste generation, which may degrade habitat quality and attract opportunistic or invasive species if not properly managed.

- Potential soil contamination from fuel or oil spills, which could impact soil structure, fertility, and soil biota.

These impacts are considered **low to moderate** and can be effectively mitigated through standard environmental management practices.

Ecosystem Modification and Habitat Loss

Habitat loss represents one of the primary ecological risks associated with development. However, the site is semi-natural and already subject to disturbance, and does not fall within irreplaceable or highly sensitive conservation areas.

Key impacts include:

- Removal of vegetation within the development footprint, resulting in localised habitat loss.
- Potential loss of individuals of protected or conservation concern species if not identified and managed prior to construction.
- Disturbance to fauna during construction activities, including displacement and temporary habitat avoidance.
- Reduction in available microhabitats for smaller fauna such as reptiles, amphibians, and invertebrates.

Given the current land use and disturbance levels, these impacts are considered **moderate**, with **low residual impact** following mitigation.

Erosion and Invasive Alien Plant Risks

Disturbance of soil and vegetation cover during construction increases the risk of erosion and invasion by alien plant species.

Key impacts include:

- Localised soil erosion, particularly in disturbed areas such as access roads and panel installation zones.
- Establishment and spread of invasive alien plant species, which may outcompete indigenous vegetation and alter ecosystem functioning.
- Long-term degradation of vegetation if invasive species are not actively managed.

These impacts are considered **moderate** but manageable through the implementation of an invasive species control plan and erosion management measures.

Hydrological and Geomorphological Processes

No permanent wetlands or major aquatic systems were identified within the project area. However, minor drainage features are present and may be affected by construction activities.

Key impacts include:

- Localised erosion and sedimentation associated with disturbed surfaces.
- Alteration of surface water flow patterns, potentially leading to runoff concentration and minor gully formation.
- Sedimentation of ephemeral drainage lines, which may affect vegetation establishment and soil stability.

These impacts are expected to be **low** and can be effectively mitigated through appropriate stormwater management and soil stabilisation measures.

28 RISK EVALUATION

The possible impacts the proposed activity could have on the receiving environment were further quantified by severity, and values were assigned to applicable criteria ratings. This will provide an understanding of the extent to which the proposed activity impacts the receiving environment. Therefore, scoring the significance of impacts assesses and predicts the environmental impacts through the evaluation of the following factors:

- Probability of the impact,
- Duration of the impact,
- Extent of the impact, and
- Magnitude of the impact.

The significance of those environmental impacts is then assessed by considering the proposed mitigation measures. The significance of the impact “without mitigation” is the prime determinant of the nature and degree of mitigation required. Each of the above impact factors has been used to assess each potential impact using ranking scales.

Table 9: Impact Significance Ratings.

SIGNIFICANCE POINTS	SIGNIFICANCE RATING
0 - 30 points	Low environmental significance
31 - 59 points	Moderate environmental significance
60 -100 points	High environmental significance

Determination of Significance – With Mitigation:

Determination of significance with mitigation refers to the foreseeable significance of the impact after the successful implementation of the necessary mitigation measures.

The Significance Rating (SR) is determined as follows:

$$\text{Significance Rating (SR)} = (\text{Intensity} + \text{Duration} + \text{Extent}) \times \text{Probability}$$

Identifying Potential Impacts **without Mitigation Measures** (WOM)

Following the assignment of the necessary weights to the respective aspects, the criteria are summed and multiplied by their assigned probabilities, resulting in a value for each impact (prior to the implementation of mitigation measures). Significance without mitigation is rated on the following scale (Table 9):

Identifying the Potential Impacts **with Mitigation Measures** (WM)

In order to gain a comprehensive understanding of the overall significance of the impact after the implementation of the mitigation measures, it will be necessary to re-evaluate the impact. The significance of mitigation thereafter is rated on the following scale (Table 10).

Table 10: Significance Rating Scales without Mitigation.

SR < 30	Low (L)	Impacts with little real effect, and which should not have an influence on or require modification of the project design or alternative mitigation. No mitigation is required.
30 < SR < 60	Medium (M)	Where it could have an influence on the decision unless it is mitigated. An impact or benefit which is sufficiently important to require management. Of moderate significance - could influence the decisions about the project if left unmanaged.
SR > 60	High (H)	The impact is significant, mitigation is critical reducing impact or risk. The resulting impact could influence the decision depending on the possible mitigation. An impact which could influence the decision about whether or not to proceed with the project.

Table 11: Significance Rating Scales with Mitigation.

SR < 30	Low (L)	The impact is mitigated to the point where it is of limited importance.
30 < SR < 60	Medium (M)	Notwithstanding the successful implementation of the mitigation measures to reduce the negative impacts to acceptable levels, the negative impact will remain significant. However, taken within the overall context of the project, the persistent impact does not constitute a fatal flaw.
SR > 60	High (H)	The impact is of major importance. Mitigation of the impact is not possible on a cost-effective basis. The impact is regarded as high importance and taken within the overall context of the project, is

		regarded as a fatal flaw. An impact regarded as high significance after mitigation could render the entire development option or entire project proposal unacceptable.
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29 IMPACT EVALUATION

The tables below (Tables 11, 12, and 13) summarize potential impacts during the Pre-construction, Construction, Operational, and Decommission phases of the project.

Table 12: Impacts Evaluation during Pre and Construction.

Pre-construction and Construction Phase of the Solar PV Facility and Associated Infrastructure										
Potential Impact	Probability		Duration		Extend		Magnitude		Significance Without Mitigation	Significance With Mitigation
	Without	With	Without	With	Without	With	Without	With		
1. Pre-construction: Poor site planning and layout resulting in unnecessary loss of semi-natural vegetation, protected species (including <i>Elaeodendron transvaalense</i>), and disruption of local ecological connectivity	4	2	3	2	3	2	4	2	48 (Moderate)	16 (Low)
2. Construction: Clearing of vegetation and site preparation leading to localised habitat loss and displacement of fauna	5	4	3	3	3	2	4	3	60 (High)	32 (Moderate)
3. Soil disturbance during construction leading to erosion, sedimentation of drainage lines,	5	3	3	3	3	2	4	3	60 (High)	24 (Low)

and increased susceptibility to invasive alien plant establishment										
4. Machinery operation and human activity causing noise, dust, and waste pollution affecting fauna and vegetation	4	2	2	1	2	1	3	1	32 (Moderate)	8 (Low)
5. Chemical spills and hydrocarbon leaks resulting in localised soil contamination and impacts on vegetation and soil biota	4	2	3	1	2	1	3	1	36 (Moderate)	8 (Low)
6. Alteration of surface drainage patterns during construction leading to localised runoff concentration, erosion, and sedimentation of ephemeral drainage lines	4	2	3	2	2	1	3	1	36 (Moderate)	12 (Low)
7. Drilling and pylon erection for the construction of the 2km-long powerline between PV facility and Baruto substation	4	2	3	2	2	1	3	1	36 (Moderate)	12 (Low)

Table 13: Impact Evaluation during Operational Phase.

Operational Phase of the Solar PV Facility and Associated Infrastructure										
Potential Impact	Probability		Duration		Extend		Magnitude		Significance Without Mitigation	Significance With Mitigation
	Without	With	Without	With	Without	With	Without	With		
1. Continued habitat transformation and restriction of faunal movement within the solar facility footprint	4	2	4	3	3	2	3	2	48 (Moderate)	24 (Low)
2. Disturbance to fauna due to maintenance activities, human presence, and occasional vehicle movement	3	2	3	2	2	1	2	1	24 (low)	8 (Low)
3. Potential collision risk or behavioural disturbance to avifauna associated with panel glare and infrastructure	3	2	3	2	2	1	3	1	30 (Moderate)	8 (Low)
4. Potential collision risk or behavioural disturbance to avifauna associated with pylons and powerline infrastructure	3	2	3	2	2	1	3	1	30 (Moderate)	8 (Low)

5. Spread of invasive alien plant species due to inadequate maintenance and disturbed soils	4	2	4	2	3	2	3	1	48 (Moderate)	16 (Low)
6. Localised soil compaction and vegetation degradation due to ongoing maintenance access	3	2	4	2	2	1	2	1	24 (Low)	8 (Low)
7. Alteration of natural vegetation structure beneath and around panels (e.g., shading effects)	3	2	4	3	2	1	2	1	24 (Low)	12 (Low)

Table 14: Impact Evaluation during Decommissioning Phase.

Decommissioning Phase of the Solar PV Facility										
Potential Impact	Probability		Duration		Extend		Magnitude		Significance Without Mitigation	Significance With Mitigation
	Without	With	Without	With	Without	With	Without	With		
1. Vegetation clearing and disturbance during the removal of infrastructure, leading to temporary habitat loss	4	2	2	1	2	1	3	1	32 (Moderate)	8 (Low)
2. Soil disturbance, compaction, and erosion associated with dismantling activities	4	2	2	1	2	1	3	1	32 (Moderate)	8 (Low)
3. Generation of waste materials (panels, cables, infrastructure) leading to potential pollution if not properly managed	4	2	2	1	2	1	3	1	32 (Low)	8 (Low)
4. Risk of chemical contamination from residual materials (e.g., oils, fuels)	3	2	2	1	2	1	2	1	24 (Low)	8 (Low)
5. Disturbance to fauna due to increased human activity and noise during dismantling	3	2	2	1	2	1	2	1	24 (Low)	8 (Low)

6. Failure to rehabilitate disturbed areas leading to long-term degradation and invasive species establishment	4	2	4	2	3	2	3	1	48 (Moderate)	16 (Low)
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30 RESIDUAL IMPACTS

The proposed Phase 1 of Gilimberg Solar PV facility at a 300ha site is not expected to result in significant long-term irreversible impacts, particularly given the site's semi-natural condition and the implementation of mitigation and rehabilitation measures. However, some **residual impacts** may persist beyond the operational and decommissioning phases (e.g., loss of habitat).

The following residual impacts are anticipated:

- **Ecological or biodiversity-related impacts**

- Localised and permanent transformation of habitat within the development footprint, resulting in a reduction in available habitat for some species.
- Potential long-term alteration of species composition, with more disturbance-tolerant species persisting in the area.
- Partial and localised reduction in ecological connectivity, although broader landscape connectivity is expected to remain intact.

- **Hydrological-related impacts**

- Minor and localised changes in surface water flow patterns due to altered land cover and infrastructure placement.
- Potential for ongoing localised erosion or sedimentation if rehabilitation is not fully effective, particularly in disturbed areas.

- **Geomorphological or landscape impacts**

- Localised and potentially persistent soil compaction in areas subjected to prolonged infrastructure placement or access routes.
- Alteration of the natural landscape character within the project footprint due to the presence of solar infrastructure.

Overall, residual impacts are expected to be **low and localised**, and largely reversible over time through appropriate rehabilitation and site management.

31 CUMULATIVE IMPACT

Cumulative impacts refer to the combined effects of the proposed development together with existing and reasonably foreseeable activities within the broader landscape.

The proposed Phase 1 of Gilimberg Solar PV facility at a 300ha site, is located within a landscape that already includes **existing infrastructure**, such as power lines and a substation, as well as ongoing land uses, including livestock grazing, resource harvesting, and small-scale sand mining.

The cumulative impacts associated with the proposed development include:

- **Ecological impacts**

- Incremental loss and fragmentation of semi-natural vegetation, which may contribute to a gradual reduction in habitat availability at a local scale.
- Potential cumulative effects on species of conservation concern, particularly if additional developments occur in the surrounding area.
- Moderate reduction in ecological connectivity at a local scale; however, the broader regional connectivity is expected to remain functional.
- **Vegetation and ecosystem function**
 - Gradual alteration of vegetation structure and composition due to combined land uses and infrastructure development.
 - Reduced ecosystem resilience in already disturbed areas, particularly where invasive alien species are not effectively controlled.
- **Hydrological and soil processes**
 - Localised increases in runoff and erosion where vegetation cover is reduced across multiple sites.
 - Potential cumulative sedimentation of ephemeral drainage lines, although impacts are expected to remain limited in extent.
- **Disturbance-related impacts**
 - Incremental increases in noise, dust, and human activity within the landscape, which may affect both wildlife and nearby land users.
 - Visual impacts associated with increased infrastructure, although these are not expected to significantly alter the broader landscape character.

32 PROPOSED IMPACT MANAGEMENT MEASURES

The proposed Phase 1 of Gilmberg Solar PV facility at a 300ha site, is situated within a **medium-sensitivity, semi-natural landscape that supports a protected species (Marula Tree) and has a high likelihood of additional Species of Conservation Concern (SCC) occurring**, including *Elaeodendron transvaalense*. The site is subject to moderate disturbance from grazing, firewood collection, and sand mining, but still retains ecological function and conservation value.

Given these conditions, **avoidance, minimisation, and protection of sensitive features (drainage lines) are essential**, supported by search and rescue of **protected trees** where avoidance is not feasible. The measures below must guide all phases of the project.

Site Planning and No-Go Areas

- The **avoidance and minimisation principles** of the Mitigation Hierarchy must be applied at all times. Infrastructure must be positioned within the least sensitive and already disturbed areas of the site (avoid all the drainage lines [Figure 11](#)).
- **Search, Rescue and Relocation** of protected and SCC species must be undertaken before site clearance. The ecologist must undertake this, execute the relocation process, and ensure their survival at their receptor site.
 - An experienced ecologist will determine the number of individuals to be relocated per species.
- The drainage lines, including their buffers, must first be receptors for the relocated plants and declared/marked and communicated as a **no-go area**.
- Buffers must be:
 - Clearly **demarcated in the field** using stakes, danger tape, or fencing
 - Communicated to all construction personnel
- All infrastructure (panels, roads, cabling) must be **micro-sited to avoid these buffers**.
- Ephemeral drainage lines must be **avoided as far as possible**, and no infrastructure should obstruct natural flow paths.

Species Protection and Search and Rescue

- A **pre-construction walkdown** must be undertaken by a qualified specialist to confirm the presence of all protected species and SCC. This must be in the wet season to maximise the identification and locating of all plants.
 - Marked and recorded all plants (species and quantity)
 - Avoided during construction
- The **“avoidance”** of the Mitigation Hierarchy is not viable in the clearance of the site (300ha), therefore:
 - Only the Marula Tree is confirmed on site, and the likelihood of *Elaeodendron transvaalense*.
 - A **search and rescue/relocation program for the SCC/protected species must be implemented** prior to site clearing (in wet season)
 - A qualified specialist must conduct all relocations.
 - Necessary **permits must be obtained** from the relevant authority for the rescue and relocation of the plants.
- Relocated individuals must be:
 - Transplanted to a suitable habitat with similar soil and vegetation conditions (prioritise the drainage lines/systems and their buffer). The experienced ecologist must advise and assist in this task.

- Monitored to assess survival and establishment. At least once a month, these replanted plants must be monitored/inspected at their receptor sites to ensure survival.
- All data (locations, relocations, mortality) must be **recorded and available for audit**.

Vegetation Clearance and Rehabilitation

- Vegetation clearing must be restricted strictly to the 300ha (approved footprint).
- Disturbance of **semi-natural vegetation must be minimised**, especially in less impacted areas and those of outside the project area.
- Topsoil must be:
 - Stripped ($\pm 150\text{--}200$ mm)
 - Stockpiled separately
 - Protected from erosion and contamination
- Topsoil must be **reused during rehabilitation** to preserve seed banks.
- Rehabilitation of the site disturbed during or by these proposed activities must:
 - Be implemented progressively where possible, as the project progresses.
 - Use **locally indigenous species** and locally endemic
 - Prioritise rapid ground cover establishment
- No collection of firewood or harvesting of indigenous vegetation by workers is permitted.

Soil Erosion and Stormwater Management

- Erosion control measures must be installed prior to and during construction, including:
 - Silt fences
 - Berms
 - Stormwater diversion channels
- Runoff must not be concentrated in a manner that causes:
 - Gully formation
 - Sedimentation of drainage lines
- Ephemeral drainage lines must:
 - Be protected from disturbance
 - Maintain natural flow patterns
- Exposed areas must be stabilised using:
 - Mulch
 - Brush packing
 - Other suitable methods
- Regular inspections must be conducted, especially after rainfall events.

Alien Invasive Plant Management

- An **Alien Invasive Plant Control Plan** must be implemented for the duration of the project.
- All invasive species must be:
 - Identified
 - Controlled or eradicated
- Disturbed areas must be monitored regularly for:
 - New infestations
 - Re-establishment of invasive species
- Control methods may include:
 - Mechanical removal
 - Chemical control (where permitted)
- Rehabilitation areas must be monitored to ensure the successful establishment of indigenous vegetation.
- **Faunal Protection and Disturbance Mitigation**
- A pre-construction survey must identify:
 - Burrows
 - Nests
 - Microhabitats
- Construction activities should be limited to **daylight hours** where possible.
- Trenches and excavations must:
 - Be inspected daily and include escape ramps where necessary
- Speed limits must be enforced to reduce collisions with fauna.
- No hunting, trapping, or disturbance of fauna by workers is permitted.
- Where feasible, undertake targeted searches for Giant Jewel Beetles (*Sternocera orissa orissa*) during peak activity periods prior to vegetation clearing.
- Where individuals are encountered, implement relocation to a suitable habitat outside the development footprint, under specialist supervision.
- Vegetation clearing should be undertaken in a phased manner to allow fauna and invertebrates to disperse naturally.
- Acknowledge that avoidance of host plant species (e.g., *Vachellia*, *Senegalia*, *Dichrostachys*, *Gymnosporia*) is not entirely feasible; however:
- These species are abundant in the surrounding areas.
- Adequate alternative habitat remains available.

Pollution Prevention and Waste Management

- Fuels and chemicals must be stored in **bunded and designated areas**.
- Spill kits must be available on site at all times.
- Any spills must be:

- Contained immediately
 - Cleaned and disposed of correctly
- Solid waste must be:
 - Collected and stored in designated areas
 - Disposed of at licensed facilities
- Dust suppression (e.g., water spraying) must be applied where necessary.
- Sanitation facilities must be provided and properly maintained.

Operational Management

- Maintenance activities must be confined to **designated access routes**.
- Vegetation clearing during operation must be minimised.
- Ongoing monitoring must include:
 - SCC presence
 - Invertebrates of cultural significance
 - Erosion
 - Invasive species
- All mitigation measures (erosion control, alien plant control) must continue throughout the operation.

Decommissioning and Rehabilitation

- Infrastructure must be removed with **minimal additional disturbance**.
- Compacted soils must be:
 - Ripped
 - Rehabilitated
- Disturbed areas must be:
 - Recontoured
 - Topsoil replaced
 - Re-vegetated with indigenous species
- Post-decommissioning monitoring must ensure:
 - Vegetation recovery
 - Stability of soils
 - Absence of invasive species

Compliance, Monitoring, and Reporting

- An independent **Environmental Control Officer (ECO)** must be appointed.
- ECO responsibilities must include:
 - Monitoring buffer compliance (15 m SCC buffers)
 - Verification of search and rescue activities
 - Erosion and stormwater inspections
 - Alien plant control monitoring

- Rehabilitation progress
- Regular compliance reports must be submitted to the Competent Authority.
- All mitigation actions must be:
 - Documented
 - Auditable
 - Adaptively managed based on monitoring results

33 CONCLUSION

The site proposed for Phase 1 of the Gilimberg Solar PV facility at a 300ha site area is situated within a semi-natural landscape that supports moderate levels of biodiversity and ecological function. The site has been historically utilised for game farming and subsistence agriculture and is currently used for livestock grazing, with additional disturbances from firewood collection and localised sand mining. Despite these pressures, the vegetation remains largely intact and continues to provide habitat and ecological connectivity at a local scale.

The assessment confirmed the presence of one protected species, and there is a high likelihood of occurrence of Species of Conservation Concern (SCC), including the Near Threatened and protected *E. transvaalense*. The site also falls within the Vulnerable Makhado Sweet Bushveld vegetation type, particularly along the powerline corridor, although portions of this vegetation have already been impacted by existing infrastructure.

The ecological sensitivity of the site is therefore considered moderate, with a Medium Present Ecological Status (PES) and Medium Ecological Importance and Sensitivity (EIS). While the site retains ecological value, it does not represent a pristine or irreplaceable system, and impacts associated with the proposed development are expected to be localised and manageable.

Potential impacts include habitat loss, disturbance to fauna and flora, soil erosion, invasive alien plant spread, and localised changes to ecological connectivity. However, these impacts are typical of solar PV developments and can be effectively mitigated.

The implementation of strict mitigation measures will reduce ecological risk to some extent. Key requirements include:

- Application of the mitigation hierarchy is recommended.
 - This prioritises the **avoidance and minimising** of impacts on all drainage lines, as shown in the [Figure 4](#) and [Figure 11](#).
- Pre – construction **botanical walk or search and rescue** to identify all the SCC locations and quantity must be undertaken (before vegetation/site clearance). These plants must be relocated and replanted.
- Micro-siting of infrastructure to avoid sensitive features (drainage lines)

- Protection of semi-natural vegetation and drainage features with at least 50m buffer
- Implementation of erosion control and stormwater management
- Establishment of an Alien Invasive Plant Control Programme and Rehabilitation Plan
- The rehabilitation of all areas destroyed during this project must prioritise the use of indigenous species
- A waste management plan that must prioritise avoidance, minimising, prevention and protection of the environment must be developed and adopted. All persons working in the plant must be educated/inducted on this. The informative and awareness boards/posters/leaflets must be clearly displayed in publicly accessible areas.
- Ongoing monitoring and compliance by an Environmental Control Officer (ECO) throughout the whole project life span.

With the above measures in place, the significance of impacts can be reduced to low after mitigation. Based on the findings of the assessment and specialist judgement, the proposed Phase 1 Gilimberg RE 1, 200MW Cluster Solar PV Facility is considered environmentally acceptable, with low residual ecological risk, provided that all mitigation measures are implemented, monitored, and enforced throughout the project lifecycle.

34 REFERENCES

- Brownlie, S., Botha, M., Helme, N., Holness, S., Rebelo, T., and Wessels, K., 2009. *Guidelines for Incorporating Biodiversity into Environmental Impact Assessment Processes*. Pretoria: South African National Biodiversity Institute (SANBI).
- Department of Environmental Affairs (DEA), 2016. *National Protected Area Expansion Strategy (NPAES)*. Pretoria: DEA.
- Driver, A., Sink, K., Nel, J., Holness, S., Van Niekerk, L., Daniels, F., Jonas, Z., Majiedt, P., Harris, L., and Maze, K., 2011. *National Biodiversity Assessment 2011: An assessment of South Africa's biodiversity and ecosystems*. Pretoria: SANBI.
- Driver, A., Sink, K.J., Nel, J.L., et al., 2012. *National Biodiversity Assessment 2011: Technical Report*. Pretoria: SANBI.
- Edwards, D., et al., 2018. *Best practice guidelines for biodiversity-inclusive impact assessment*. Pretoria: SANBI.
- Friedmann, Y. and Daly, B. (eds.), 2004. *Red Data Book of the Mammals of South Africa: A Conservation Assessment*. Pretoria: CBSG Southern Africa.
- Mucina, L. and Rutherford, M.C. (eds.), 2006. *The Vegetation of South Africa, Lesotho and Swaziland*. Pretoria: SANBI.
- Nel, J.L. and Driver, A., 2014. *National Biodiversity Assessment: Freshwater Component*. Pretoria: SANBI.
- Republic of South Africa, 1996. *Constitution of the Republic of South Africa (Act No. 108 of 1996)*. Pretoria: Government Printer.
- Republic of South Africa, 1998. *National Environmental Management Act (Act No. 107 of 1998)*. Pretoria: Government Printer.
- Republic of South Africa, 1998. *National Water Act (Act No. 36 of 1998)*. Pretoria: Government Printer.
- Republic of South Africa, 2004. *National Environmental Management: Biodiversity Act (Act No. 10 of 2004)*. Pretoria: Government Printer.
- SANBI (South African National Biodiversity Institute), 2010. *Red List of South African Plants*. Pretoria: SANBI.
- SANBI, 2019. *National Biodiversity Assessment 2018: Synthesis Report*. Pretoria: SANBI.
- SANBI, 2024. *Protected Areas Database*. Pretoria: South African National Biodiversity Institute.
- Skinner, J.D. and Chimimba, C.T., 2005. *The Mammals of the Southern African Subregion*. Cambridge: Cambridge University Press.

35 APPENDICES

35.1 Appendix A: Historical records of flora

N O.	Scientific name	Common name	SANBI REDLIST	PROTECTED SPECIES
1	<i>Abutilon austroafricanum</i>		LC	
2	<i>Achyranthes aspera</i>	Devil's Horsewhip	Exotic	
3	<i>Acokanthera oppositifolia</i>	bushmans poison	LC	
4	<i>Acokanthera rotundata</i>	Round-leaf Poison-bush	LC	
5	<i>Acrotome inflata</i>	Horsefright	LC	
6	<i>Aloe greatheadii</i>	spotted aloe	LC	
7	<i>Aloe marlothii</i>	mountain aloe	LC	
8	<i>Ammocharis coranica</i>	Ground lily	LC	
9	<i>Antizoma angustifolia</i>		LC	
10	<i>Apodytes dimidiata</i>	White Pear	LC	
11	<i>Aptosimum lineare</i>	Bushveld Carpet Flower	LC	
12	<i>Asparagus virgatus</i>	African Broom Asparagus	LC	
13	<i>Blepharis subvolubilis</i>	Eye Lashes	LC	
14	<i>Bolboschoenus glaucus</i>	tuberous bulrush	LC	
15	<i>Boophone disticha</i>	Common Oxbane	LC	
16	<i>Brachylaena rotundata</i>	Mountain Silver-oak	LC	
17	<i>Buddleja saligna</i>	False Olive	LC	
18	<i>Carissa bispinosa</i>	num-num	LC	
19	<i>Carissa spinarum</i>	Climbing Numnum	LC	
20	<i>Cleome maculata</i>	Spotted Spindlepod	LC	
21	<i>Cleome monophylla</i>	Singleleaf Spindlepod	LC	
22	<i>Cleome rubella</i>	Pretty Lady	LC	
23	<i>Combretum molle</i>	Fluweelboswilg	LC	
24	<i>Crassula sarcocaulis</i>	Bonsai Stonecrop	LC	
25	<i>Cryptolepis cryptolepioides</i>		LC	
26	<i>Cucumis metuliferus</i>	Jelly Melon / Rooikomkommer	LC	
27	<i>Datura ferox</i>	White Stinkweed	1b	
28	<i>Dicerocaryum senecioides</i>	Savanna Devilthorn	LC	
29	<i>Dichrostachys cinerea</i>	sicklebush	LC	
30	<i>Dicoma tomentosa</i>	Woolly Karmedik	LC	
31	<i>Diospyros lycioides</i>	Quilted Bluebush	LC	
32	<i>Dombeya rotundifolia</i>	African Wild Pear	LC	
33	<i>Drimia altissima</i>	Tall Squill	LC	
34	<i>Ehretia rigida</i>	Puzzle Bush	LC	
35	<i>Elaeodendron transvaalense</i>	anthill saffron	NT	Protected tree
36	<i>Eragrostis superba</i>	Heartseed Lovegrass	LC	
37	<i>Euclea crispa</i>	Blue Guarri	LC	
38	<i>Euclea undulata</i>	Common Guarri	LC	
39	<i>Euphorbia cooperi</i>	Candelabra Naboom	LC	

40	<i>Euphorbia ingens</i>	Common Naboom	LC	
41	<i>Euphorbia tirucalli</i>	Rubber Milkbush	LC	
42	<i>Ficus abutilifolia</i>	Large-leaved rock fig	LC	
43	<i>Ficus burkei</i>	Common Wild Fig	LC	
44	<i>Flaveria bidentis</i>	speedyweed	1b	
45	<i>Gomphocarpus tomentosus</i>	Common Woolly Milkweed	LC	
46	<i>Gossypium herbaceum africanum</i>	Wild Cotton	LC	
47	<i>Grewia flavescens</i>	Sandpaper Raisin	LC	
48	<i>Grewia retinervis</i>	Kalahari Raisin	LC	
49	<i>Gymnosporia buxifolia</i>	Common Spikethorn	LC	
50	<i>Gymnosporia senegalensis</i>	confetti spikethorn	LC	
51	<i>Heteromorpha arborescens</i>	mMakatlala	LC	
52	<i>Heteropyxis natalensis</i>	Lavender-tree	LC	
53	<i>Hypoestes forskalii</i>	White Ribbon Flower	LC	
54	<i>Kalanchoe paniculata</i>		LC	
55	<i>Kalanchoe sexangularis</i>	Bushveld Kalanchoe	LC	
56	<i>Kleinia longiflora</i>	paintbrush flower	LC	
57	<i>Ledebouria luteola</i>	highveld African hyacinth	LC	
58	<i>Leucas capensis</i>	White Lionspaw	LC	
59	<i>Leucas martinicensis</i>	White Tumbleweed	LC	
60	<i>Lippia javanica</i>	Fever Tea	LC	
61	<i>Lopholaena coriifolia</i>	Small-leaved Fluff-bush	LC	
62	<i>Maerua afra</i>	Bush Maerua	LC	
63	<i>Melhaniania prostrata</i>	Common Melhaniania	LC	
64	<i>Melia azedarach</i>	Syringa	1b	
65	<i>Mystroxydon aethiopicum</i>	Kooboo-berry	LC	
66	<i>Olea europaea africana</i>	African Wild Olive	LC	
67	<i>Opuntia ficus-indica</i>	Prickly Pear Cactus	1b	
68	<i>Osteospermum muricatum</i>	Buchu Boneseed	LC	
69	<i>Osyris lanceolata</i>	Rock Tannin-bush	LC	
70	<i>Pappea capensis</i>	Jacket plum	LC	
71	<i>Pellaea calomelanos</i>		LC	
72	<i>Peltophorum africanum</i>	Weeping wattle	LC	
73	<i>Persicaria decipiens</i>	slender knotweed	LC	
74	<i>Philyrophyllum schinzii</i>		LC	
75	<i>Plumbago auriculata</i>	blue plumbago	LC	
76	<i>Pogonarthria squarrosa</i>	Herringbone Grass	LC	
77	<i>Pollichia campestris</i>	Waxberry Plant	LC	
78	<i>Pterocarpus rotundifolius</i>	Round-leaved Bloodwood	LC	
79	<i>Salvia leucantha</i>	Mexican Bush Sage	LC	
80	<i>Sansevieria aethiopica</i>	Mother-in-law's Tongue	LC	
81	<i>Sansevieria trifasciata</i>	Snake Plant	LC	
82	<i>Schinus molle</i>	Peruvian Pepper Tree	Invasive (not listed)	

83	<i>Schotia brachypetala</i>	Weeping Boerbean	LC	
84	<i>Sclerocarya birrea afra</i>	Marula	LC	Protected Tree
85	<i>Searsia lancea</i>	Karee	LC	
86	<i>Searsia leptodictya</i>	Mountain Karree	LC	
87	<i>Searsia pyroides</i>	Common Wild Currant	LC	
88	<i>Searsia tumulicola</i>	Hardleaf Curranthrus	LC	
89	<i>Senecio barbertonicus</i>	Succulent Bush Ragwort	LC	
90	<i>Senecio pleistocephalus</i>		LC	
91	<i>Senegalia afra</i>	Common Hook-thorn Acacia	LC	
92	<i>Solanum campylacanthum</i>	bitter-apple	LC	
93	<i>Tagetes minuta</i>	Tall Khaki Weed	LC	
94	<i>Tarchonanthus camphoratus</i>	Wild Camphorbush	LC	
95	<i>Tecoma stans</i>	yellow bells	1b	
96	<i>Tetradenia brevispicata</i>	Small Gingerbush	LC	
97	<i>Tetradenia riparia</i>	River Gingerbush	LC	
98	<i>Themeda triandra</i>	Red Grass	LC	
99	<i>Vachellia hebeclada</i>	candlepod thorn	LC	
100	<i>Vachellia karroo</i>	Sweetthorn	LC	
101	<i>Vachellia nilotica</i>	Egyptian acacia	LC	
102	<i>Vachellia robusta</i>	moOka	LC	
103	<i>Vachellia tortilis</i>	umbrella thorn	LC	
104	<i>Vangueria infausta</i>	wild-medlar	LC	
105	<i>Vangueria parvifolia</i>	Bergwildemispel	LC	
106	<i>Viscum rotundifolium</i>	Redberry Mistletoe	LC	
107	<i>Volkameria glabra</i>	Cat's-whiskers	LC	
108	<i>Xanthium strumarium</i>	rough cocklebur	1b	
109	<i>Xenostegia tridentata</i>	Narrowleaf Merremia	LC	
110	<i>Zantedeschia albomaculata</i>	Spotted Arum	LC	
111	<i>Ziziphus mucronata</i>	buffalo-thorn	LC	

35.2 Appendix B: Historical records of fauna

No.	Scientific name	Common name	IUCN STATUS	PROTECTED SPECIES
Mammals				
1	<i>Aepyceros melampus</i>	Impala	LC	
2	<i>Atelerix frontalis frontalis</i>	Southern Hedgehog	LC	LEMA protected
3	<i>Chlorocebus pygerythrus</i>	Vervet Monkey	LC	
4	<i>Civettictis civetta</i>	African Civet	LC	LEMA protected
5	<i>Galago moholi</i>	Lesser Bushbaby	LC	
6	<i>Helogale parvula parvula</i>	Southern Dwarf Mongoose	LC	
7	<i>Herpestes sanguineus</i>	Common Slender Mongoose	LC	
8	<i>Heterohyrax brucei granti</i>	Limpopo Yellow Spotted Dassie	LC	LEMA protected
9	<i>Hystrix africaeaustralis</i>	Porcupine	LC	
10	<i>Lepus saxatilis</i>	Scrub Hare	LC	
11	<i>Lupulella mesomelas</i>	Black-backed Jackal	LC	
12	<i>Mungos mungo</i>	Banded Mongoose	LC	
13	<i>Papio ursinus griseipes</i>	Grayfoot Chacma Baboon	LC	
14	<i>Paraxerus cepapi</i>	Smith's Bush Squirrel	LC	
15	<i>Pedetes capensis</i>	Springhare	LC	
16	<i>Phacochoerus africanus</i>	Common Warthog	LC	
17	<i>Procavia capensis</i>	Rock Dassie	LC	
18	<i>Raphicerus campestris</i>	Steenbok	LC	LEMA protected
19	<i>Sylvicapra grimmia</i>	Common Duiker	LC	
Amphibians and Reptiles				
1	<i>Acanthocercus atricollis</i>	Southern Tree Agama	LC	LEMA protected
2	<i>Afrotyphlops schlegelii</i>	Schlegel's Beaked blind snake	LC	
3	<i>Agama aculeata</i>	Ground Agama	LC	LEMA protected
4	<i>Amietia delalandii</i>	Common River Frog	LC	
5	<i>Aparallactus capensis</i>	Cape Centipede-Eater	LC	
6	<i>Chamaeleo dilepis</i>	Flap-necked Chameleon	LC	LEMA protected
7	<i>Chiromantis xerampelina</i>	Grey Foam-nest Tree Frog	LC	
8	<i>Chondrodactylus turneri</i>	Turner's Thick-toed Gecko	LC	LEMA protected
9	<i>Dasypeltis scabra</i>	Egg-eating Snake	LC	
10	<i>Homopholis wahlbergii</i>	Wahlberg's Velvet Gecko	LC	LEMA protected

11	<i>Matobosaurus validus</i>	Common Giant Plated Lizard	LC	LEMA protected
12	<i>Merolus squamulosus</i>	Gewone Skurwe Akkedis	LC	LEMA protected
13	<i>Myriopholis longicauda</i>	Long-tailed Thread Snake	LC	
14	<i>Naja mossambica</i>	Mozambique Spitting Cobra	LC	
15	<i>Pachydactylus capensis</i>	Cape Thick-toed Gecko	LC	LEMA protected
16	<i>Pachydactylus punctatus</i>	Pointed Thick-toed Gecko	LC	LEMA protected
17	<i>Philothamnus semivariatus</i>	Spotted Bush Snake	LC	
18	<i>Phrynomantis bifasciatus</i>	Red-Banded Rubber Frog	LC	
19	<i>Poyntonophrynus fenoulheti</i>	Fenoulhet's Pygmy Toad	LC	
20	<i>Pyxicephalus edulis</i>	Edible bullfrog	LC	
21	<i>Rhinotyphlops lalandei</i>	Delalande's Beaked Blind Snake	LC	
22	<i>Schismaderma carens</i>	Red Toad	LC	
23	<i>Sclerophrys garmani</i>	Garman's Toad	LC	
24	<i>Stigmochelys pardalis</i>	Leopard Tortoise	LC	LEMA protected
25	<i>Trachylepis margaritifer</i>	Rainbow Skink	LC	LEMA protected
26	<i>Trachylepis punctatissima</i>	Speckled Rock Skink	LC	LEMA protected
27	<i>Trachylepis varia</i>	Variable Skink	LC	LEMA protected
28	<i>Varanus niloticus</i>	Nile Monitor	LC	LEMA protected
29	<i>Xenopus laevis</i>	Common Platanna	LC	
Invertebrates				
1	<i>Acraea neobule</i>	Wandering Donkey Acraea	LC	
2	<i>Anax ephippiger</i>	Vagrant Emperor	LC	
3	<i>Anoplolepis custodiens</i>	Pugnacious Ant	Not listed	
4	<i>Anthia burchelli</i>	Burchell's Oogpister	Not listed	
5	<i>Anthia omoplata</i>	Twospot Ground Beetle	Not listed	
6	<i>Argiope australis</i>	Common Garden Orbweb Spider	Not listed	
7	<i>Augacephalus junodi</i>	Golden Baboon Spider	Not listed	
8	<i>Belenois aurota</i>	Pioneer White	LC	
9	<i>Brachythemis leucosticta</i>	Banded Groundling	LC	
10	<i>Catantops melanostictus melanostictus</i>		Not listed	
11	<i>Chrysopoloma isabellina</i>		Not listed	
12	<i>Clinteroides permutans</i>	Larger Variable Fruit Chafer	Not listed	
13	<i>Crocothemis erythraea</i>	Broad Scarlet	LC	
14	<i>Crocothemis sanguinolenta</i>	Small Scarlet	LC	
15	<i>Cyligramma latona</i>	Creamstriped Owl	Not listed	
16	<i>Cypholoba notata</i>		Not listed	

17	<i>Cyrtacanthacris tatarica</i>	Brown-spotted Locust	Not listed	
18	<i>Deroplax silphoides</i>	Common Voodoo Shieldbug	Not listed	
19	<i>Eremoides bicristatus</i>		Not listed	
20	<i>Gabunia ruficeps</i>		Not listed	
21	<i>Gomphidia quarrei</i>	Quarre's Tiger	LC	
22	<i>Grammodes euclidioides</i>	Yellow Lines	Not listed	
23	<i>Graphipterus wahlbergi</i>		Not listed	
24	<i>Graphipterus westwoodi</i>		Not listed	
25	<i>Hadogenes newlandsi</i>		Not listed	
26	<i>Hodotermes mossambicus</i>	Rysmier	Not listed	
27	<i>Hycleus oculatus</i>	CMR Blisterbeetle	Not listed	
28	<i>Junonia hierta cebrene</i>	African Yellow Pansy	LC	
29	<i>Leucocelis amethystina</i>	Amethyst Small Fruit Chafer	Not listed	
30	<i>Locris arithmetica</i>	Spotted Red Spittlebug	Not listed	
31	<i>Miomantis caffra</i>	South African Mantis	Not listed	
32	<i>Munza basimacula</i>	Small Broadborder Cicada	Not listed	
33	<i>Mylothris agathina</i>	Eastern Dotted Border	LC	
34	<i>Oryctes boas</i>	Boas Rhinoceros Beetle	Not listed	
35	<i>Petascelis remipes</i>	Giant Twig Wilter	Not listed	
36	<i>Phoxomela umbrosa</i>		Not listed	
37	<i>Phymateus viridipes</i>	Green Milkweed Locust	Not listed	
38	<i>Pinacopteryx eriphia</i>	Zebra White	LC	
39	<i>Platycorynus dejeani</i>	Metallic Milkweed Leaf Beetle	Not listed	
40	<i>Polistes badius</i>		Not listed	
41	<i>Polyclaeis longicornis</i>	Gaudy Weevil	Not listed	
42	<i>Polyrhachis schistacea</i>	Savanna Spiny Sugar Ant	Not listed	
43	<i>Sphingomorpha chlorea</i>	Sundowner moth	Not listed	
44	<i>Stephenia caldarena</i>	Black Tip Acraea	Not listed	
45	<i>Sternocera orissa orissa</i>	Giant Jewel Beetle	Not listed	
46	<i>Synclera traducalis</i>	Variegated Pearl	Not listed	
47	<i>Telchinia serena</i>	Dancing Amber	LC	
48	<i>Tephraea morosa</i>		Not listed	
49	<i>Uroplectes carinatus</i>	Common Lesser-Thicktail Scorpion	Not listed	
50	<i>Uroplectes triangulifer</i>	Highveld Lesser-Thicktail Scorpion	Not listed	
51	<i>Utetheisa pulchella</i>	Crimson-speckle Footman	Not listed	

35.3 Appendix C: Photos from the site



